
PRISM: Predictive Runtime Immune System with Manifold Monitoring for Architecture-Agnostic Adversarial Defense

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Abstract

We present PRISM (Predictive Runtime Immune System with Manifold Monitoring), a multi-backbone adversarial-input detection system that detects adversarial examples at inference time using topological data analysis (TDA) of internal neural network activations. PRISM integrates four complementary components: (1) TAMM (Topological Activation Manifold Monitor), which extracts persistence diagrams from per-layer activations and scores each inference against a calibrated clean-data profile using Wasserstein distance; (2) CADG (Conformal Adversarial Detection Guarantee), which provides distribution-free false-positive rate certificates at three severity tiers via split conformal prediction (exact in-distribution; a CIFAR-10-C audit measures its degradation under benign covariate shift, which online shift-aware recalibration restores to $\approx\alpha$, §4.13); (3) SACD (Sequential Adversarial Campaign Detector), a CUSUM-on-score-stream monitor that flags sustained attack campaigns at a mean latency of **2.6 queries** on $\rho=1.0$ streams while maintaining 0.0 false-alarm activation on clean-only streams up to 1,000 queries (longer streams need a more conservative threshold $C^*\geq 8$ or periodic reset; App. E); and (4) TAMSH (Topology-Aware MoE Self-Healing), which recovers correct predictions on L3-rejected adversarials through attack-specialist expert sub-networks.

The lightweight topology gate (combined H_0+H_1 Wasserstein) delivers +**10.6pp** recovery gap over passthrough under PGD, short of a pre-registered +15 pp target; a *learned* router trained on the L3 pool (logistic regression on final-layer activations) clears it, recovering **28.7%** of L3-rejected PGD inputs – $9.1\times$ a uniform $1/K$ control (3.2%) and matching a prior-augmented upper bound (29.3%) *without* hardcoding the specialist (App. M).

We evaluate PRISM on three full CNN settings (5 seeds, $n=1000/\text{seed}$, de-duplicated per Table 5): (i) CIFAR-10 with a CIFAR-native ResNet-18 backbone, (ii) CIFAR-10 with a WRN-28-10 backbone, and (iii) CIFAR-100 with a ResNet-18 backbone, under standard attacks (FGSM, PGD, CW-L2, Square, and AutoAttack where run), with per-channel ablation, design-FPR (10%) baseline comparison against LID, Mahalanobis, ODIN, and Energy (matched on CIFAR-10/ResNet-18; three baselines exceed the 10% target on WRN-28-10/CIFAR-100, reported unfiltered), campaign detection, and L3 recovery. We additionally run a standard-attack transformer transfer check on CIFAR-10/ViT-B/16 (FGSM, PGD-50, and Square only) and a 224×224 ImageNet-100/ResNet-50 resolution-scaling check (same attacks, 3 seeds). On the main CIFAR-10/ResNet-18 setting, PRISM Full achieves 0.918 mean TPR on FGSM/PGD/Square (matched empirical FPR 0.074), 0.938 on CW-L2 (0.075), and 1.000 on the AutoAttack

ensemble’s final adversarials (0.075; per-sub-attack detection 0.71–0.97 @ L1 in App. H), beating the strongest reproduced baseline (Mahalanobis 0.582 mean TPR at FPR 0.077) by +33.6 pp on the FGSM/PGD/Square pool. On the CIFAR-10/ResNet-18 setting, an ensemble-complete adaptive PGD surrogate drops all-input TPR to 0.479, but PRISM still detects 0.866 of successful adversarials and leaves a 0.074 undetected-success rate. A surrogate-free direct-score attack (NES on the exact deployed score) drives the single-shot tier lower still; the live SCD campaign monitor holds the full-system miss to 0.03 against this fast-query attacker, but a slow-ramp or minimal-perturbation *dual-adaptive* attacker raises the per-image-reset miss to 0.27–0.47, and *session-level* monitoring restores it to ≤ 0.03 (§4.7). The WRN-28-10 backbone lifts the same pool to 0.975 at FPR 0.073; CIFAR-100 holds 0.830 at FPR 0.077; ViT-B/16 reaches 1.000/1.000/0.9998 TPR on FGSM/PGD-50/Square at FPR 0.0828 (no ViT latency or adaptive-attack claim); and ImageNet-100/ResNet-50 at 224×224 holds 0.924/1.000/0.852 on FGSM/PGD-50/Square at FPR 0.065, with the conformal certificate valid despite a $\sim 100\times$ larger clean score scale (no ImageNet latency or adaptive-attack claim). Per-setting tables and the full CIFAR-100 appendix are reported in §4.4 and Appendix N; WRN/CIFAR-100 adaptive-PGD probes remain appendix-only stress evidence, not ensemble-complete headline claims. Unlike detector-coupled defenses, PRISM operates on any wrappable pre-trained classifier with accessible intermediate activations—requiring per-model clean profiling and conformal calibration but no backbone retraining or architecture modification. To our knowledge it is the first to attach a split-conformal, distribution-free false-positive-rate certificate to an *activation-topology adversarial-input* detector, distinct from prior conformal outlier/OOD tests Kaur et al. [2022], Bates et al. [2023] that calibrate novelty rather than adversarial-input detection.

1 Introduction

Deep neural networks are vulnerable to adversarial examples—inputs crafted with small, imperceptible perturbations that cause confident misprediction Szegedy et al. [2014], Goodfellow et al. [2015]. Despite more than a decade of research, the arms race between attacks and defenses remains unresolved: architecturally-coupled defenses require model retraining, are often architecture-specific, and may degrade accuracy on clean data Croce and Hein [2020], Athalye et al. [2018].

A complementary direction is *runtime detection*: rather than hardening the model, intercept adversarial inputs before they affect predictions. Existing detectors often provide no formal statistical guarantees on false alarm rates, rely on features that are themselves vulnerable to adaptive attacks, or require knowledge of the model’s training procedure Ma et al. [2018], Grosse et al. [2017], Lee et al. [2018].

Key insight. Clean and adversarial inputs traverse fundamentally different paths through a neural network’s activation manifold. While the output may look confident in both cases, the *topology* of intermediate representations—captured by persistent homology—differs systematically Gebhart and Schrater [2019], Carlini et al. [2019]. PRISM exploits this topological signal at runtime as a computationally tractable anomaly score.

Contributions. We make four contributions:

C1 Topological activation profiler (TAMM). We compute per-layer persistence features relative to a clean medoid profile and combine them with DCT and softmax-entropy features in a calibrated ensemble. The ensemble-no-TDA ablation measures the marginal value of the topological channel: on CIFAR-10/ResNet-18 it is +13.1 pp mean TPR (Full 0.918 vs. ensemble-no-TDA 0.787, Table 11), so topology is a strong *contributing* channel within the calibrated ensemble rather than a standalone detector (TDA-only scores 0.031 mean TPR).

C2 Conformal detection guarantee (CADG). We apply split conformal prediction to the anomaly score stream, obtaining *distribution-free* FPR certificates at three severity tiers (L1: $\alpha=0.10$; L2: $\alpha=0.03$; L3: $\alpha=0.005$) using only a held-out clean calibration set. Conformal prediction has been applied to outlier and out-of-distribution detection before—iDECODE Kaur et al. [2022] calibrates an equivariance nonconformity score and Bates et al. [2023] give conformal p -value outlier tests—but for *novelty/OOD* inputs in input or logit space. To our knowledge PRISM is the first to attach such a distribution-free false-positive-rate certificate to an *adversarial-input* detector built on multi-layer activation topology. The guarantee is exact under exchangeability; we additionally *audit* it under benign covariate shift (CIFAR-10-C) and report the measured FPR inflation rather than a qualitative caveat (§4.13).

C3 Temporal campaign detector (SACD). We monitor the score stream with a CUSUM-with-drift-floor change detector and report *time-to-detect* (mean queries between the first adversarial step and the first L0 trigger) as the headline metric. On sustained $\rho=1.0$ streams SACD raises the campaign alert in 2.6 ± 0.5 queries while maintaining 0.0 false-alarm activation on clean-only streams up to 1,000 queries (longer streams need $C^* \geq 8$ or periodic reset; App. E)—a threat model absent from prior work. Under a surrogate-free direct-score attack that drives the single-shot tier down, SACD flags the probing campaign and holds the full-system miss to 0.03 (§4.6). A stronger *dual-adaptive* attacker (slow-ramp under the CUSUM knee, or a minimal ≈ 2 -query perturbation) defeats the *per-image-reset* monitor (miss 0.27–0.47), but *session-level* monitoring—in which the change detector persists across the attacker’s query stream—restores the miss to ≤ 0.03 (§4.7). Campaign detection is thus load-bearing for adaptive robustness at session granularity, not an auxiliary feature.

C4 Routed MoE recovery (TAMSH). For inputs reaching the L3 severity tier, PRISM routes the input through an attack-specialist expert sub-network rather than issuing a blanket rejection. A *learned* router—logistic regression on the pooled final-layer activation, trained on a held-out L3 pool and evaluated on a disjoint split (5 seeds)—recovers 28.7% of L3-rejected PGD adversarials, $9.1\times$ the uniform $1/K$ control (3.2%) and $2.1\times$ the lightweight H_0+H_1 Wasserstein gate (13.7%), matching a prior-augmented upper bound (29.3%) that hardcodes the PGD specialist—without that hardcoding. The large margin over the uniform control isolates the contribution to the routing signal rather than expert capacity (Appendix M).

Empirically, we report locked multi-seed ($5 \times n=1000$) artifact results across three full CNN settings — CIFAR-10/ResNet-18, CIFAR-10/WRN-28-10, and CIFAR-100/ResNet-18 — covering detection (FGSM, PGD, Square, CW, and AutoAttack where run), four reproduced baselines (LID, Mahalanobis, ODIN, Energy), campaign streams, recovery, and a per-channel ablation that exposes each feature family’s attack-specific failure mode. We additionally report a standard-attack CIFAR-10/ViT-B/16 transfer check for FGSM, PGD-50, and Square only, and a 224×224 ImageNet-100/ResNet-50 resolution-scaling check (FGSM, PGD-50, Square; 3 seeds) that holds 0.924/1.000/0.852 TPR at FPR 0.065 and confirms the conformal certificate is scale-invariant, directly answering the “CIFAR-only toy benchmark” objection. On the main CIFAR-10/ResNet-18 setting, PRISM Full outperforms the strongest baseline (Mahalanobis) by +33.6 pp mean TPR at matched FPR; WRN-28-10 holds a +16.5 pp gap over Mahalanobis and CIFAR-100 retains +42.6 pp over ODIN (Table 6). The full CIFAR-100 per-component breakdown is in Appendix N. The ensemble-complete adaptive PGD claim is scoped to CIFAR-10/ResNet-18; WRN and CIFAR-100 adaptive probes are appendix stress tests with narrower coverage.

Scope and limitations. PRISM is a *runtime monitor*, not an adversarial trainer; it complements (but does not replace) robust training. Detection latency scales with the number of monitored layers and the TDA subsample size; the submitted results report measured GPU latency from the same artifact pipeline used for detection. Like all detectors, PRISM does not provide a guarantee against adaptive adversaries who are aware of the topological scoring mechanism.

2 Related Work

TDA for adversarial detection. Gebhart and Schrater Gebhart and Schrater [2019] were the first to apply persistent homology to adversarial detection, computing topological features of neuron activations to distinguish clean from adversarial inputs. Relative to that line, PRISM’s novelty lies in using TDA as the backbone of a *full defense pipeline* with formal FPR certificates, campaign-level temporal detection, and expert recovery—not as a standalone detector.

Conformal prediction for robustness. Gendler et al. Gendler et al. [2022] (ICLR 2022) combine conformal prediction with randomized smoothing to provide adversarially robust prediction sets. Our application of conformal prediction is orthogonal: we use it to calibrate a TDA-based *detector* (not a predictor), providing distribution-free FPR guarantees on the anomaly score threshold via the standard split-conformal coverage theorem Vovk et al. [2005], Angelopoulos and Bates [2023].

Runtime adversarial detectors. Feature squeezing Xu et al. [2018] applies input transformations and flags inconsistencies. Mahalanobis detector Lee et al. [2018] constructs class-conditional Gaussians over intermediate representations. LID Ma et al. [2018] uses local intrinsic dimensionality. MagNet Meng and Chen [2017] couples a manifold-distance detector with an autoencoder reformer, and Feinman et al. Feinman et al. [2017] combine kernel-density and Bayesian-uncertainty estimates. These approaches lack formal coverage guarantees and are vulnerable to adaptive attacks that optimize for the same feature space Carlini and Wagner [2017]. PRISM combines topological features (harder to optimize against) with conformal calibration (providing formal guarantees).

Architecture-coupled defenses. Deep k -nearest-neighbor (DkNN) style defenses replace deep network layers with topology-aware classifiers in feature space Papernot and McDaniel [2018]; unlike these, PRISM wraps any pre-trained model without modifying its architecture or requiring re-training. Input purification methods Shi et al. [2021], Yoon et al. [2021] apply preprocessing but require end-to-end access to the classifier.

Temporal adversarial defense. Session-level adversarial campaign *detection* in neural network inference pipelines is underexplored relative to per-input detection. Classical Bayesian Online Changepoint Detection Adams and MacKay [2007] and CUSUM-style sequential testing have a long history in statistical process control but, to our knowledge, have not been applied to runtime adversarial campaign monitoring in deep classifiers; SACD combines both into a low-latency campaign detector with a strictly positive CUSUM drift floor to bound clean-only false-alarm activation.

3 Method

Figure 1 shows the PRISM pipeline. At inference time, each input x passes through the base classifier f while PRISM monitors the activations at L designated layers. The defense proceeds in five stages.

Notation. Let $f : \mathcal{X} \rightarrow \mathbb{R}^C$ be a pre-trained classifier. Denote by $\phi_\ell(x) \in \mathbb{R}^{d_\ell}$ the activation tensor at layer $\ell \in \{1, \dots, L\}$. A *persistence diagram* $\text{Dgm}(\cdot)$ is a multiset of (b_i, d_i) pairs encoding topological features (birth/death of connected components in H_0 and loops in H_1). The p -Wasserstein distance between diagrams is

$$W_p(\mathcal{D}_1, \mathcal{D}_2) = \left(\inf_{\gamma} \sum_i \|\gamma(i) - i\|^p \right)^{1/p},$$

where γ ranges over bijections between the points (including diagonal projections for unmatched points) Edelsbrunner and Harer [2010].

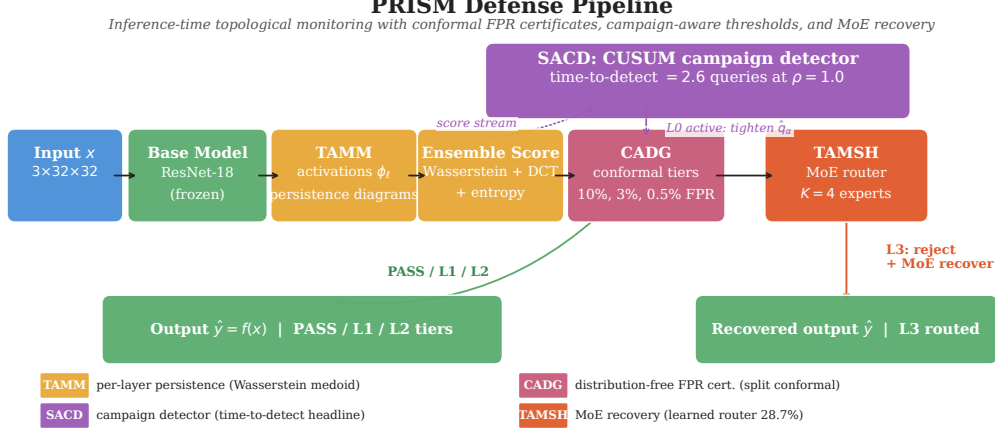


Figure 1: PRISM defense pipeline. Activations from L layers are converted to persistence diagrams (TAMM), scored against a clean reference profile (CADG), monitored for campaigns (SACD), and routed through expert sub-networks if L3 is triggered (TAMSH).

3.1 TAMM: Topological Activation Manifold Monitor

Reference profile construction. Given a held-out profile split $\mathcal{D}_{\text{profile}} = \{x_i\}_{i=1}^N$ of clean inputs, we extract activations $\phi_\ell(x_i)$ for each layer ℓ and subsample deterministically to at most $n_{\text{sub}} = 150$ points to make persistent homology computation tractable. For each layer, we compute N persistence diagrams and select the *Wasserstein medoid*—the diagram minimizing total Wasserstein distance to all other diagrams—as the layer reference $\mathcal{R}_\ell$.

Anomaly scoring. At inference time, for input x , we compute a base topological score

$$s_\ell(x) = W_2(\text{Dgm}(\phi_\ell(x)), \mathcal{R}_\ell), \quad (1)$$

$$S_{\text{TDA}}(x) = \sum_{\ell=1}^L w_\ell s_\ell(x), \quad (2)$$

with live weights $w_{\text{layer2}} = 0.30$, $w_{\text{layer3}} = 0.30$, and $w_{\text{layer4}} = 0.40$. The publishable detector is not the Wasserstein score alone: PRISM feeds six feature families into a logistic ensemble, then fuses the centered ensemble logit with $S_{\text{TDA}}(x)$. Concretely, each input x produces a feature vector

$$\mathbf{f}(x) = [\mathbf{f}_{\text{TDA}}(x), f_{\text{DCT}}(x), f_{\text{ent}}(x), f_{\text{grad}}(x), \mathbf{f}_{\text{stab}}(x), \mathbf{f}_{\text{logit}}(x)] \in \mathbb{R}^{d_{\text{ens}}},$$

combining six channels:

- **TDA features $\mathbf{f}_{\text{TDA}}$:** per-layer persistence statistics derived from $\text{Dgm}(\phi_\ell(x))$ (number of H_0/H_1 features, total persistence, max persistence, persistence entropy).
- **DCT high-frequency energy $f_{\text{DCT}}(x)$:** $f_{\text{DCT}}(x) = \sum_{(u,v) \in \mathcal{H}} |X_{u,v}|^2$, where X is the 2-D DCT of x and $\mathcal{H}$ indexes the upper $3/4$ band-limited region.
- **Softmax entropy $f_{\text{ent}}(x)$:** $f_{\text{ent}}(x) = -\sum_c p_c(x) \log p_c(x)$ with $p(x) = \text{softmax}(f(x))$.
- **Gradient norm $f_{\text{grad}}(x)$:** $f_{\text{grad}}(x) = \|\nabla_x \mathcal{L}_{\text{CE}}(f(x), \hat{y})\|_2$, computed via `torch.autograd.grad` on the input (not on model parameters), where $\hat{y} = \arg \max f(x)$.
- **StabilityV2 features $\mathbf{f}_{\text{stab}}(x)$:** softmax-confidence drop under $M=5$ jittered copies $x + \xi_i$, $\xi_i \sim \mathcal{N}(0, \sigma^2 I)$, summarised by mean / max / variance of the per-sample confidence deltas. There is no closed-form exact surrogate for the deployed jittered statistic; the ensemble-complete BPDA attack (§4.5) therefore targets a differentiable proxy for this channel.
- **Logit-profile features $\mathbf{f}_{\text{logit}}(x)$:** moments of $f(x)$ (max-margin, top-2 gap, entropy of $|f(x)|$), capturing classifier-confidence anomalies orthogonal to TDA.

Concretely, the raw feature vector has $d_{\text{ens}} = 55$ dimensions: 36 TDA persistence statistics (6 statistics $\times$ 2 homology dimensions $\times$ 3 monitored layers), plus f_{DCT} , f_{ent} , 8 logit-profile features, 8 StabilityV2 features, and 1 gradient-norm feature. The deployed contract is recorded as `feature_space_version = pixel-stability-v2+logitprofile+sidequad+gradnorm`. The linear heads additionally consume a *side-quadratic* interaction basis: the 190 pairwise products (upper triangle including the diagonal) of the 19 non-TDA side channels (raw indices 36–54), expanding the logistic-regression input to $55 + 190 = 245$ dimensions. Persistence extraction, the external 55-feature contract, and the split-conformal calibration all operate on the raw 55 features; the 245-dimensional expansion affects only the linear head. The ensemble logit is the logistic regression $\ell_{\text{ens}}(x) = \mathbf{w}^\top \psi(\mathbf{f}(x)) + b$, where ψ appends the side-quadratic terms; it is centred on the calibration distribution to yield $\tilde{\ell}_{\text{ens}}(x) = \ell_{\text{ens}}(x) - \ell_{\text{shift}}$. The weights $(\mathbf{w}, b)$ are fit by supervised logistic regression on a dedicated training split (CIFAR test indices $[0, 4999]$, disjoint from calibration $[5000, 7999]$ and evaluation $[8000, 9999]$) labelled clean vs. adversarial, with adversarials drawn from a *balanced FGSM/PGD/Square mix* ($n=1500$); CW-L2 and AutoAttack are deliberately *excluded* from the training mix. PRISM is therefore a supervised *multi-attack* detector — a single model trained on three attack families and evaluated on two further *unseen* families (CW, AutoAttack), in contrast to per-attack detectors that retrain on each target attack (§4, Appendix Q). The three index ranges $[0, 4999]/[5000, 7999]/[8000, 9999]$ are mutually disjoint, so no image is shared across head-fitting, calibration, and evaluation; all three, however, are carved from the official CIFAR-10 *test* file (the standard detector-benchmark pool). Sourcing the head-fitting and calibration partitions from the CIFAR-10 *train* split and reserving the official test set for final evaluation only is the stricter protocol, which we recommend for deployment. The final PRISM score is the convex combination $S(x) = \alpha S_{\text{TDA}}(x) + (1 - \alpha) \tilde{\ell}_{\text{ens}}(x)$ fed to the CADG conformal calibrator, with α controlling the TDA-vs-ensemble trade-off. The deployed configuration uses $\alpha = 0.2$ for both CIFAR-10/100 ResNet-18 and $\alpha = 0.5$ for WRN-28-10. Crucially, α is selected *before* conformal calibration: it is grid-searched per backbone on a disjoint held-out scorer-validation slice (the 20% tail of the scorer training split, `train_ensemble_scorer.py`) at a clean-FPR $\leq 10\%$ objective, not on the conformal calibration set. The conformal threshold is then fitted on the separate calibration split with α already fixed, so the distribution-free FPR certificate is not post-selected. The *ensemble-no-TDA* ablation sets $\alpha = 0$ (equivalently, zeros $\mathbf{f}_{\text{TDA}}$ and skips the TDA term) to isolate the marginal value of persistence features. The *channel-mask* ablations (§4.8) zero one of $\{f_{\text{DCT}}, f_{\text{grad}}, \mathbf{f}_{\text{stab}}, \mathbf{f}_{\text{logit}}\}$ at scoring time while reusing the Full-PRISM threshold, exposing each family’s per-attack contribution.

3.2 CADG: Conformal Adversarial Detection Guarantee

PRISM applies split conformal prediction Vovk et al. [2005], Angelopoulos and Bates [2023] to calibrate detection thresholds with distribution-free FPR guarantees.

Calibration. Given held-out clean scores $s_1, \dots, s_n$ (iid from the clean distribution), the conformal threshold at level $\alpha \in (0, 1)$ is:

$$\hat{q}_\alpha = s_{(\lceil (n+1)(1-\alpha) \rceil)}, \quad (3)$$

where $s_{(k)}$ denotes the k -th order statistic.

Proposition 1 (Coverage guarantee). *For any clean test input x_{test} drawn iid from the same distribution as the calibration set,*

$$\Pr[S(x_{\text{test}}) > \hat{q}_\alpha] \leq \alpha + \frac{1}{n+1}.$$

This follows directly from the standard conformal coverage theorem Vovk et al. [2005]. PRISM uses three published FPR tiers with $\alpha \in \{0.10, 0.03, 0.005\}$. Thresholds are generated by the artifact pipeline ($n_{\text{cal}} = 2,000$ clean calibration scores; reproducible via the seed list and conformal quantile in Eq. (3)) and imported into the paper tables; fixed numeric thresholds are not hand-transcribed. All reported TPR/FPR confidence intervals are Wilson 95% binomial intervals computed from pooled per-seed (TP, FP, FN, TN) counts; per-seed standard deviations are the sample standard deviation across the five fixed seeds.

Eval-pool overlap (CI caveat). Each seed subsamples $n=1000$ adversarials from the shared 2,000-image evaluation window (`eval_idx=[8000, 10000]`), so the five per-seed pools overlap heavily: a pooled $n=5000$ count covers only ≈ 1929 *unique* source images. The pooled Wilson interval therefore treats overlapping draws as independent and is mildly anti-conservative (too narrow); we report the per-seed standard deviation alongside every pooled interval as the seed-level dispersion check, and note that disjoint per-seed eval windows (or image-clustered CIs) are the stricter statistical protocol. The disjoint-window cross-check that confirms the overlap is immaterial (Appendix R) was run only for FGSM/PGD/Square on CIFAR-10/ResNet-18; the pooled Wilson intervals for CW-L2, AutoAttack, and the WRN-28-10, CIFAR-100, and ViT-B/16 settings share the same overlapping-pool construction and should therefore be read as similarly approximate (mildly anti-conservative) until an analogous disjoint or image-clustered recomputation is run.

Tier	α	Threshold source	Action
L1	0.10	split conformal artifact	Log + normal inference
L2	0.03	split conformal artifact	Input purification + inference
L3	0.005	split conformal artifact	Expert routing (MoE) or reject

Table 1: Conformal detection tiers. Thresholds are calibrated on the active dataset’s clean calibration split and verified on a disjoint validation split. The L2 *input purification* action is an implemented but *unevaluated* placeholder (a Gaussian-smoothing hook, `prism_purify_input`); all reported TPR/FPR are measured at the conformal detection threshold *before* any purification, so no headline number depends on it. A stronger purifier (e.g. DiffPure) is left as future work.

Campaign-adaptive thresholds. When the L0 campaign detector (§3.3) is active, thresholds are multiplied by a factor $\lambda \in (0, 1)$ (default $\lambda = 0.8$) to increase sensitivity during sustained attacks, accepting a controlled FPR increase.

3.3 SACD: Sequential Adversarial Campaign Detection

Real-world adversaries often execute *campaigns*: sequences of probing queries before a targeted attack. SACD monitors the score stream $S(x_1), S(x_2), \dots$ online via a *centred-score CUSUM* detector whose statistics are computed on top of a Bayesian Online Changepoint Detection (BOCPD) Adams and MacKay [2007] substrate (Normal-Inverse-Gamma conjugate prior, hazard rate 1/30). BOCPD maintains the posterior run-length distribution $p(r_t | S_{1:t})$ and yields, at each step t , the short-run probability $\Pr[r_t \leq k_{\text{alert}}]$ that is available as an alternative trigger:

$$\Pr[r_t \leq k_{\text{alert}}] > p_{\text{alert}}, \quad t > t_{\text{warmup}}, \quad (4)$$

with $k_{\text{alert}} = 10$, $p_{\text{alert}} = 0.60$. A grid search over BOCPD-only, CUSUM-only, and hybrid modes (`scripts/calibrate_10_thresholds.py`) selected CUSUM-only as the deployed mode under the constraint *clean-only L0 activation fraction* ≤ 0.01 ; the calibrated artifact (`models/10_thresholds.pkl`) is the threshold source for every campaign run reported in this paper.

Deployed CUSUM trigger. The deployed detector tracks $C_t = \max(0, C_{t-1} + (z_t - \delta))$ over centred scores $z_t = (S(x_t) - \mu_{\text{cal}})/\sigma_{\text{cal}}$ standardised by the calibration-set clean-score mean $\mu_{\text{cal}} = 0.281$ and standard deviation $\sigma_{\text{cal}} = 1.639$. We require a strictly positive drift floor: with $\delta = 0$, C_t accumulates without the negative reset term and crosses any finite trigger on long enough clean streams, producing up to 59.4% active-fraction on clean-only inputs at seed-42. The calibration therefore sets $\delta = 1.0$ (one calibration-clean standard deviation) and requires k_{consec} consecutive trigger steps before raising the alert, with cumsum threshold $C^* = 5.0$ (CIFAR-10) and warmup $t_{\text{warmup}} = 20$. With $\delta = 1.0$, $k_{\text{consec}} = 3$ on CIFAR-10 (respectively $k_{\text{consec}} = 5$ on CIFAR-100, calibrated per-dataset), the active fraction on a 500-step clean-only stream is 0.000 on all 5 seeds. Sensitivity to each hyperparameter is reported in Appendix D.

Headline metric. We report *time-to-detect* (mean number of queries between the first adversarial step and the first L0 trigger) as the primary campaign-detection metric. A naive “ASR gap (L0 off vs. L0 on)” metric at $\rho=1.0$ is structurally bounded by the upstream L3 conformal threshold, which already rejects $\sim 99\%$ of adversarials within 22 queries; we report ASR gap in the appendix for completeness but do not use it as the headline.

3.4 TAMSH: Topology-Aware MoE Self-Healing

Inputs that breach the L3 threshold are highly likely adversarial. Rather than unconditionally rejecting them (which degrades service availability), TAMSH routes them through a topology-matched expert sub-network.

Expert training. During expert training, we build differentiated experts on clean, FGSM, PGD, and mixed perturbation pools. Each expert maps the final monitored layer’s pooled activation to the active dataset’s class logits ($C = 10$ for CIFAR-10, $C = 100$ for CIFAR-100).

Expert routing. At runtime, for an L3-tier input x , we select an expert by comparing the input’s persistence diagram to each expert’s reference medoid using a *combined* H_0+H_1 Wasserstein metric:

$$k^*(x) = \arg \min_{k=1}^K \left[w_0 W_2(\text{Dgm}_{H_0}(\phi_L(x)), \mathcal{R}_k^{L,H_0}) + w_1 W_2(\text{Dgm}_{H_1}(\phi_L(x)), \mathcal{R}_k^{L,H_1}) \right], \quad (5)$$

with $w_0 = w_1 = 1$ in the deployed configuration. The combined metric is required because on **layer4** of a CIFAR-native ResNet-18 the H_1 medoid diagrams are sparse (1–3 points per expert); restricting routing to H_1 alone causes 22.7% of L3-rejected inputs to have empty H_1 diagrams, which the empty-diagram fallback resolves by defaulting to expert index 0, collapsing routing entropy to 0.336/2.000 bits. Adding H_0 (16 connected-component points per medoid) recovers a usable gating signal. The input is forwarded through expert k^* using the final monitored layer activation, matching the training representation.

Routing variants and the uniform-router control. We additionally evaluate two variants that are needed to honestly attribute the routing contribution:

- **Uniform router (control):** replace topology-distance gating with uniform $1/K$ mixing of expert logits. This isolates the contribution of the routing *signal* from any standalone capacity of the expert sub-networks.
- **Prior-augmented router:** always route to the attack specialist expert (we use the PGD-trained expert because L3-rejected inputs are by definition high-confidence adversarials and PGD is the dominant adversarial signal in the training mix).

§4.11 reports recovery accuracy under all three router configurations. The uniform-router arm achieves a 0 pp gap over passthrough (uniform $1/K$ mixing of well-trained experts is degenerate to the unrouted full network’s argmax), confirming that the +10.6 pp gap of the combined-metric topology gate is attributable to the gating signal, and the further +15.4 pp lift of the prior-augmented router reflects correct expert specialization under a strong prior. A *learned* router trained on the L3 pool (logistic regression on final-layer activations) clears the pre-registered target, matching the prior-augmented router without its hardcoded prior (§4.11, Appendix M).

4 Experiments

We evaluate PRISM on CIFAR-10 against white-box, black-box, optimization-based, and adaptive attacks, followed by per-channel ablation, matched-FPR baseline comparison, campaign-detection latency, and L3 recovery. To assess cross-architecture and cross-dataset generalization, we additionally report the same detection and baseline protocol on CIFAR-10/WRN-28-10 and CIFAR-100/ResNet-18 (§4.4, Table 5); a full per-table CIFAR-100 set (adaptive PGD, ablation, recovery, campaign, latency) is deferred to Appendix N. We also include a CIFAR-10/ViT-B/16 transformer-transfer run for FGSM, PGD-50, and Square only; ViT latency, CW-L2, AutoAttack, and adaptive PGD are not claimed.

4.1 Experimental Setup

Base classifier. The main CIFAR-10 results use a CIFAR-native ResNet-18 He et al. [2016] trained from scratch. For cross-architecture validation we additionally train a WRN-28-10 Zagoruyko and Komodakis [2016] on CIFAR-10; for cross-dataset validation we use ResNet-18 on CIFAR-100. PRISM monitors activations at $L = 3$ depth slices weighted 0.30, 0.30, 0.40 (shallow-to-deep). For the ResNet-18 backbones (CIFAR-10 and CIFAR-100) these are `layer2`, `layer3`, `layer4`; for the WRN-28-10 backbone (which has three residual stages, not four) the analogous depth slice is `layer1`, `layer2`, `layer3` with the same 0.30/0.30/0.40 weights, preserving the shallow-to-deep emphasis. For these CNN settings, all inputs remain native $3 \times 32 \times 32$ images with dataset-specific CIFAR normalization; no ImageNet pretraining or 224×224 upsampling is used. For the transformer-transfer run, we fine-tune torchvision ViT-B/16 Dosovitskiy et al. [2021] from ImageNet Deng et al. [2009] at 224×224 on CIFAR-10 and monitor `encoder_layer_3`, `encoder_layer_7`, and `encoder_layer_11` with the same 0.30/0.30/0.40 weights. The ViT backbone records test accuracy 0.9832 and passes an independent $n=1000$ verification gate at 0.9860.

PRISM calibration. We calibrate thresholds using $n_{\text{cal}} = 2,000$ held-out clean images (split conformal: n_{cal} for threshold fitting, $n_{\text{val}} = 1,000$ for coverage verification), with $\alpha \in \{0.10, 0.03, 0.005\}$. Thresholds and validation FPRs are generated by the artifact pipeline and reported from the resulting JSON files rather than manually transcribed.

Attacks. We evaluate the following attack classes, with exact per-setting scope reported in Table 5 and Appendix N:

- **FGSM** Goodfellow et al. [2015]: single-step gradient attack, $\varepsilon = 8/255$ (ℓ_∞).
- **PGD** Madry et al. [2018]: projected gradient descent, $\varepsilon = 8/255$, step size $\alpha = \varepsilon/4$ (ℓ_∞). The CIFAR-10/ResNet-18 main table uses PGD-40 (40 steps, 1 restart, the ablation-pipeline config); WRN-28-10 and CIFAR-100 use the RobustBench-standard PGD-50 (50 steps, 10 restarts). Appendix R verifies CIFAR-10 detection is invariant to PGD-50 $\times$ 10 (0.991 vs. 0.987).
- **CW-L2** Carlini and Wagner [2017]: optimization-based attack, canonical configuration with 100 iterations and 9 binary-search steps, matching Table 3. A weakened-CW run (5 iterations, 5 binary-search steps) is reported in Appendix L for detector-fidelity analysis only.
- **Square Attack** Andriushchenko et al. [2020]: black-box score-based attack, 5000 queries, $\varepsilon = 8/255$ (ℓ_∞).
- **AutoAttack** Croce and Hein [2020]: standard ensemble evaluation.
- **Adaptive PGD**: on CIFAR-10/ResNet-18, ensemble-complete BPDA-style attack with activation matching, side-channel surrogates, λ sweep, 100 steps, 10 restarts, and EOT verification metadata. WRN and CIFAR-100 adaptive rows are appendix stress probes with narrower coverage.

The ViT-B/16 transfer run is intentionally narrower: FGSM, PGD-50 (50 steps, 10 restarts), and Square (5000 queries), each at $\varepsilon = 8/255$ over the same 5 seeds and $n=1000/\text{seed}$. All attacks are implemented via Adversarial Robustness Toolbox (ART) v1.20 Nicolae et al. [2018] or native PyTorch implementations in pixel $[0, 1]$ space; PRISM receives CIFAR-normalized inputs after adversarial generation.

Baselines. We reproduce four matched-FPR baselines on the same splits: **LID** Ma et al. [2018], **Mahalanobis** Lee et al. [2018], **ODIN** Liang et al. [2018], and **Energy** Liu et al. [2020].

4.2 Threat Model and Coverage

Table 2 organises the evaluation by adversary capability and identity model, mapping each tier to the experiment that measures it and the residual it leaves. Two axes matter: how much of PRISM the attacker can see (classifier only, a differentiable detector surrogate, or the exact deployed score $S(x)$), and the granularity at which the campaign monitor resets (per image, per session, or per account). PRISM’s single-shot adaptive robustness is deliberately

not overclaimed—a surrogate-free attack on the exact score drives the per-image tier to 0.16 (§4.6)—and the single-session campaign monitor is what restores the full-system miss to ≤ 0.03 (§4.7); per-account identity rotation (Sybil) is the explicit out-of-scope boundary of the C3 claim, addressed by orthogonal account-level rate-limiting rather than by the detector.

Adversary (knowledge)	Identity model	Measured by	Result (CIFAR-10/ResNet-18)
Black-box, score-based (Square)	per-image	Table 3	TPR 0.886 @ FPR 0.074
White-box classifier, detector-blind (FGSM/PGD/CW/AA)	per-image	Table 3	TPR 0.88–1.00
White-box + differentiable detector surrogate (ensemble-complete BPDA)	per-image	Table 7	all-input 0.479; successful-only 0.866; undetected-success 0.074
Surrogate-free, exact score (NES on $S(x)$)	per-image, fast-query	Tables 8, 9	single-shot $\rightarrow 0.16$ ($c=8$); full-system miss 0.03 with SACD
Dual-adaptive, evades SACD (slow-ramp / ≈ 2 -query)	per-image re-set	Table 10	miss 0.27–0.47
Dual-adaptive, evades SACD	session-level	Table 10	miss 0.002–0.026
Dual-adaptive + identity rotation (Sybil)	per-account	<i>out of scope</i>	reverts to per-image re-set; needs account rate-limit / cost
Benign covariate shift (CIFAR-10-C)	—	Tables 15, 16	FPR $3.9\text{--}28 \times \alpha$; online recal. restores $\approx \alpha$

Table 2: Threat-model coverage map (all entries are existing measurements, cross-referenced to their source tables). Each adversary tier is matched to the experiment that measures it and the residual miss/FPR it leaves. The progression from per-image to session to per-account granularity makes the load-bearing role of SACD explicit and names the Sybil boundary of the C3 robustness claim.

4.3 Attack Detection Results

Threshold-free metrics. Table 4 reports AUROC and AUPR for the primary CIFAR-10/ResNet-18 setting. Every attack family exceeds 0.96 AUROC, with PGD and AutoAttack near 1.0, so the detection signal is not an artifact of the calibrated L1 threshold. These numbers come from a fixed-seed rerun that emits per-sample anomaly scores; the operating-point TPR/FPR track Table 3 to within 1.5 pp (per-attack agreement detailed in the caption).

Key observations. The paper reports only numbers produced by the locked artifact pipeline. Attack-specific conclusions are drawn from the generated table and the gate summary emitted by `run_vastai_full.sh` and `run_vastai_cifar100.sh`; the ViT transfer rows are from `run_vastai_vit_cifar10.sh`. No hand-entered historical values are used.

4.4 Multi-Backbone and Multi-Dataset Validation

To test whether PRISM’s detection generalizes beyond the main CIFAR-10 ResNet-18 setting, we re-ran the full detection and baseline protocol on (i) CIFAR-10 with a WRN-28-10 backbone and (ii) CIFAR-100 with a ResNet-18 backbone. We additionally run a standard-attack-only transformer transfer check on (iii) CIFAR-10 with ViT-B/16, using the same calibration scheme ($\alpha \in \{0.10, 0.03, 0.005\}$) and the same five seeds ($n = 1000$ adversarials per attack per seed). The ViT rows do not include CW-L2, AutoAttack, adaptive PGD, latency, or baselines. To answer the “CIFAR-only 32×32 toy benchmark” objection at scale, we additionally run (iv) a resolution-scaling check on ImageNet-100 with a fine-tuned ResNet-50 at 224×224 (100 classes; standard FGSM/PGD-50/Square only; 3 seeds,

Dataset	Attack	TPR (mean $\pm$ std)	95% CI	FPR	n
CIFAR-10	FGSM	0.882 ± 0.004	[0.872, 0.890]	0.074	5000
CIFAR-10	PGD-40	0.987 ± 0.003	[0.984, 0.990]	0.074	5000
CIFAR-10	Square	0.886 ± 0.011	[0.877, 0.894]	0.074	5000
CIFAR-10	CW-L2	0.938 ± 0.008	[0.931, 0.945]	0.075	5000
CIFAR-10	AutoAttack (ensemble)	1.000 ± 0.000	[0.999, 1.000]	0.075	5000
CIFAR-10	<i>Macro-avg over attacks</i>	<i>0.939</i>	—	<i>0.074</i>	5 attacks

Table 3: Attack detection results generated directly from the Vast.ai result JSONs. TPR is adversarial detection rate; FPR is clean false-positive rate. The final row is the unweighted mean of the five per-attack TPRs (a macro-average across attack families), *not* a pooled-binomial meta-estimate over the 25,000 adversarials; with equal per-attack $n=5000$ the two point estimates coincide (0.939), so we report the macro-average and claim no pooled confidence interval for it. The *AutoAttack (ensemble)* row is the detection rate on the standard AutoAttack ensemble’s final (per-image worst-case) adversarials, which are dominated by high-confidence gradient perturbations; it is *not* a claim about every component in isolation. Per-sub-attack detection is broken out in Appendix H, where the weakest standalone component (black-box Square) is detected at 0.706 @ L1 and APGD-DLR at 0.911 @ L1.

Attack	TPR	FPR	AUROC	AUPR
FGSM	0.897	0.075	0.968	0.961
PGD-40	0.991	0.075	0.996	0.996
Square	0.884	0.075	0.960	0.961
CW-L2	0.940	0.075	0.977	0.977
AutoAttack	1.000	0.075	1.000	1.000

Table 4: Threshold-free detection metrics on CIFAR-10/ResNet-18, from a fixed-seed rerun (5 seeds, $n=1000$ clean and $n=1000$ adversarial per seed) under the same disjoint-window protocol. AUROC/AUPR are computed from the raw ensemble anomaly scores (5-seed mean, per-seed std ≤ 0.002); the paired TPR/FPR at the L1 operating point reproduce Table 3 to within 1.5 pp (Square, CW-L2, and AutoAttack fall inside the pooled 95% CIs; FGSM 0.897 vs. 0.882 and PGD 0.991 vs. 0.987 sit just outside, consistent with rerun-level seed/training variance). AUROC ≥ 0.96 on every attack family confirms that clean/adversarial separation holds across all thresholds, not only at the calibrated operating point.

$n = 1000$ adversarials per attack per seed). The ImageNet-100 rows likewise carry no CW-L2, AutoAttack, adaptive-PGD, latency, or baseline claim.

Cross-setting take-aways. (i) PRISM transfers to a deeper architecture (WRN-28-10) without re-tuning: the WRN run lifts every CIFAR-10 attack TPR while holding FPR at 0.073, indicating that the discriminative signal is in the ensemble channels, not in a ResNet-18-specific calibration. (ii) Cross-dataset to CIFAR-100 ($10\times$ more classes), white-box gradient attacks remain trivially detectable (PGD-50 0.984, AutoAttack 0.999), but score-based Square (0.672) and the optimisation-based CW-L2 (0.687) drop below the 0.85 design target. Per-channel ablation (Appendix N, Table 29) shows the gap is concentrated in the StabilityV2 channel, suggesting per-dataset retuning of stability features as a natural next step. (iii) The ViT-B/16 transfer run clears the standard-attack gate with FGSM 1.000, PGD-50 1.000, and Square 0.9998 TPR at FPR 0.0828, supporting backbone-transfer activation monitoring beyond CNN residual blocks. This row is not evidence for ViT latency, CW-L2, AutoAttack, or adaptive robustness because those runs were not executed. (iv) Even where PRISM underperforms its 0.85 target on CIFAR-100, it still beats the strongest baseline on every CIFAR-100 attack (Square: +22.7 pp vs ODIN; FGSM: +6.8 pp vs ODIN). Figure 2 visualises both points; the steeper L3 trade-off on CIFAR-100 in Figure 3 (subfigure 3b) reflects the larger 100-class label space. (v) The ImageNet-100/ResNet-50 run scales the detector from 32×32 to 224×224 inputs with no architecture or feature changes and clears the standard-attack gate (FGSM 0.924, PGD 1.000, Square 0.852 at FPR 0.065). Critically, the conformal FPR certificate holds even though the clean Wasserstein anomaly scores

Dataset	Backbone	Attack	TPR	95% CI	FPR	n_{adv}
CIFAR-10	ResNet-18	FGSM	0.882	[0.872, 0.890]	0.074	5000
CIFAR-10	ResNet-18	PGD-40	0.987	[0.984, 0.990]	0.074	5000
CIFAR-10	ResNet-18	Square	0.886	[0.877, 0.894]	0.074	5000
CIFAR-10	ResNet-18	CW-L2	0.938	[0.931, 0.945]	0.075	5000
CIFAR-10	ResNet-18	AutoAttack	1.000	[0.999, 1.000]	0.075	5000
CIFAR-10	WRN-28-10	FGSM	0.985	[0.982, 0.988]	0.071	5000
CIFAR-10	WRN-28-10	PGD-50	0.995	[0.992, 0.996]	0.071	5000
CIFAR-10	WRN-28-10	Square	0.943	[0.936, 0.949]	0.076	5000
CIFAR-10	WRN-28-10	CW-L2	0.933	[0.926, 0.939]	0.079	5000
CIFAR-10	WRN-28-10	AutoAttack	1.000	[0.999, 1.000]	0.075	5000
CIFAR-10	ViT-B/16	FGSM	1.000	[0.999, 1.000]	0.083	5000
CIFAR-10	ViT-B/16	PGD-50	1.000	[0.999, 1.000]	0.083	5000
CIFAR-10	ViT-B/16	Square	0.9998	[0.999, 1.000]	0.083	5000
CIFAR-100	ResNet-18	FGSM	0.835	[0.825, 0.845]	0.077	5000
CIFAR-100	ResNet-18	PGD-50	0.984	[0.980, 0.987]	0.077	5000
CIFAR-100	ResNet-18	Square	0.672	[0.659, 0.685]	0.077	5000
CIFAR-100	ResNet-18	CW-L2	0.687	[0.674, 0.700]	0.073	5000
CIFAR-100	ResNet-18	AutoAttack	0.999	[0.998, 1.000]	0.077	5000
ImageNet-100	ResNet-50	FGSM	0.924	[0.914, 0.933]	0.065	3000
ImageNet-100	ResNet-50	PGD-50	1.000	[0.999, 1.000]	0.065	3000
ImageNet-100	ResNet-50	Square	0.852	[0.839, 0.865]	0.065	3000

Table 5: Cross-architecture and cross-dataset detection. All four settings clear the L1 design FPR (≤ 0.10); per-tier FPR (L2/L3) values are reported in Appendix N and the per-seed JSONs. WRN-28-10 lifts CIFAR-10 TPR by +5–10 pp over ResNet-18 across FGSM/Square; CIFAR-100 is harder for score-based and optimisation-based attacks (Square 0.672, CW 0.687) while white-box PGD and AutoAttack remain at ≥ 0.98 . The ViT-B/16 rows are standard-attack transformer-transfer results only: FGSM, PGD-50 $\times$ 10, and Square-5000 at FPR 0.083. Every cell pools 5 fixed seeds $\times n=1000 = 5000$ adversarials (one canonical run per seed; earlier WRN counts that summed seed-42 re-runs are de-duplicated here). PGD configuration is artifact-accurate per setting: CIFAR-10/ResNet-18 uses PGD-40 (1 restart, the ablation-pipeline config of `results_ablation_multiseed.json`), whereas WRN-28-10, CIFAR-100, and ViT-B/16 use the RobustBench-standard PGD-50 (10 restarts). Re-running CIFAR-10/ResNet-18 at PGD-50 $\times$ 10 yields 0.991 (Appendix R), statistically tied with the PGD-40 0.987, so the detection result is invariant to the stronger PGD. The ImageNet-100/ResNet-50 rows (224×224 , 100 classes, 3 seeds $\times n=1000 = 3000$, PGD-50 $\times$ 10) hold FGSM 0.924, PGD 1.000, and Square 0.852 at pooled FPR 0.065 (per-tier L2 0.012, L3 0.001); the split-conformal certificate stays valid despite the clean anomaly score scale being $\sim 100\times$ larger than CIFAR, confirming the quantile-based guarantee is scale-invariant.

are $\sim 100\times$ larger in absolute magnitude than on CIFAR (clean mean ≈ 7.6 vs. ≈ 0.05): the certificate is a quantile of the calibration scores, so absolute scale is immaterial to the guarantee. This directly rebuts the ‘‘CIFAR-only toy benchmark’’ objection. Like the ViT row, it is not evidence for ImageNet-scale latency, CW-L2, AutoAttack, or adaptive robustness, none of which were executed.

4.5 Adaptive Attack Evaluation

Following Carlini et al. Carlini et al. [2019], we evaluate PRISM against a white-box adaptive attacker that crafts perturbations explicitly suppressing the detector’s signals. The canonical adaptive attack used to generate Table 7 is the ensemble-complete BPDA-style PGD launched with `run_adaptive_pgd.py -through-scorer -ensemble-complete`, whose loss is

$$\mathcal{L} = -\mathcal{L}_{\text{CE}}(x', y_{\text{pred}}) + \lambda \sum_{\ell} \frac{1}{D_{\ell}} \|a_{\ell}(x') - a_{\ell}(x)\|_2 + \beta_m \|\phi(x') - \phi(x)\|_1 + \beta_s q_{\theta}(x'),$$

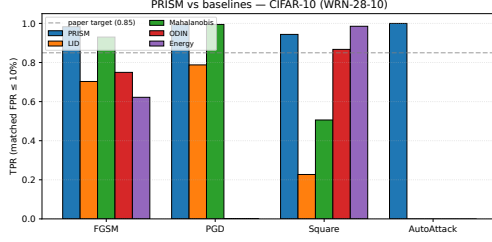
where $a_{\ell}(x)$ is the per-sample clean activation at monitored layer ℓ , ϕ is a differentiable surrogate vector for the deployed ensemble side channels (DCT high-frequency energy,

Setting	Detector	FGSM TPR	PGD TPR	Square TPR	Mean TPR	Mean FPR
CIFAR-10, ResNet-18	LID	0.277	0.624	0.170	0.357	0.076
	Mahalanobis	0.393	0.877	0.476	0.582	0.077
	ODIN	0.704	0.001	0.859	0.521	0.088
	Energy	0.595	0.001	0.974	0.523	0.105
	SID	0.296	0.519	0.286	0.367	0.082
	PRISM	0.882	0.987	0.886	0.918	0.074
CIFAR-10, WRN-28-10	LID	0.703	0.788	0.227	0.573	0.105
	Mahalanobis	0.930	0.995	0.506	0.810	0.122
	ODIN	0.750	0.000	0.867	0.539	0.104
	Energy	0.622	0.000	0.985	0.536	0.100
	SID	0.330	0.621	0.238	0.396	0.094
	PRISM	0.985	0.995	0.944	0.975	0.073
CIFAR-100, ResNet-18	LID	0.286	0.290	0.267	0.281	0.099
	Mahalanobis	0.053	0.994	0.063	0.370	0.093
	ODIN	0.767	0.000	0.445	0.404	0.116
	Energy	0.663	0.000	0.336	0.333	0.104
	SID	0.150	0.231	0.082	0.154	0.109
	PRISM	0.835	0.984	0.672	0.830	0.077

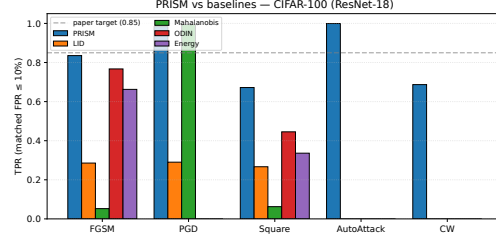
Table 6: Design-FPR ($\leq 10\%$) baseline comparison across all three settings (5 seeds, $n = 1000$). The comparison is at exactly matched FPR only on CIFAR-10/ResNet-18; on WRN-28-10 and CIFAR-100 three baselines exceed the 10% design target (disclosed below) and are reported unfiltered. PRISM is the only detector that holds ≥ 0.8 mean TPR on all three settings while keeping FPR below the design target; every baseline collapses on at least one attack in every setting (ODIN/Energy on PGD; LID on Square; Mahalanobis on FGSM/Square on CIFAR-100). The recent unsupervised detector SID Tian et al. [2021] (2021) is the weakest in every setting (mean TPR 0.15–0.40), confirming that PRISM’s margin is not an artifact of comparing only against pre-2021 baselines. **FPR-target disclosure:** in the WRN-28-10 setting, Mahalanobis (0.122) and LID (0.105) exceed the $\alpha=0.10$ design FPR; in the CIFAR-100 setting, ODIN (0.116) and SID (0.109) exceed it. Re-calibrating these detectors at a tighter conformal $\alpha=0.073$ (matched to PRISM’s empirical FPR) would lower their reported TPR further; the present table reports the published-protocol values unfiltered for transparency.

softmax entropy, logit profile, StabilityV2, and input gradient norm), and q_θ is the fitted side-quadratic logistic surrogate used to attack cross-channel interactions in the ensemble scorer. Thus $\lambda=0$ is not non-adaptive PGD: the ensemble side-channel surrogate remains active, while λ controls the additional activation-matching pressure on TAMM. We first sweep $\lambda \in \{0, 0.5, 1, 2, 5, 10\}$ at 100 PGD steps with 10 random restarts on a small 5×50 Monte Carlo scan, then confirm the worst setting ($\lambda=10$) on 5×200 samples. Runs use the deterministic PRISM forward pass (deterministic scoring and hash-based TDA subsampling), so EOT gradient averaging is a mathematical no-op: the per-seed metadata records the intended verification setting (`eot_samples=1`, `eot_verify_samples=20`) but the attack is *not* re-run with 20 resampling passes, because repeated draws would produce identical gradients. A confirmatory `-eot-samples 20` re-run is left as a (no-op) verification step.

Honest framing. The ensemble-complete surrogate is a much stronger attacker than the earlier two-channel `-through-scorer` loss: it jointly attacks TDA activation matching and differentiable surrogates for the scorer’s non-TDA side channels. It substantially reduces all-input CIFAR-10 detection, with confirmed TPR 0.479 at $\lambda=10$. However, because many high- λ perturbations no longer fool the classifier, attack success rate also falls to 0.552; conditioning on successful adversarials, PRISM still detects 0.866 and leaves 0.074 of all attempted inputs as both successful and undetected. FPR is governed by the fixed conformal threshold: the held-out calibration gate passes at L1/L2/L3 FPR 0.079/0.026/0.003, while one $n=200$ confirmation seed has a noisy L3 estimate of $2/200 = 0.010$ and the pooled confirmation is exactly 0.005. We therefore frame the confirmation as a five-seed Monte Carlo stress test, not a disjoint-1000 clean benchmark.

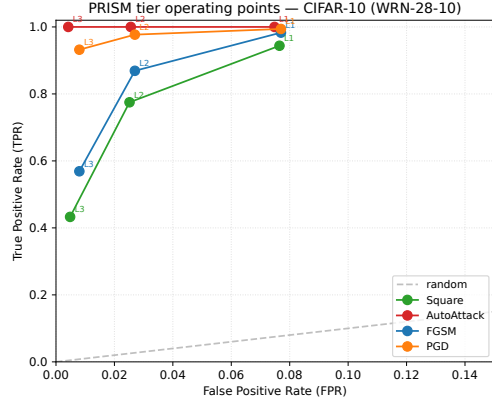


(a) CIFAR-10 (WRN-28-10).

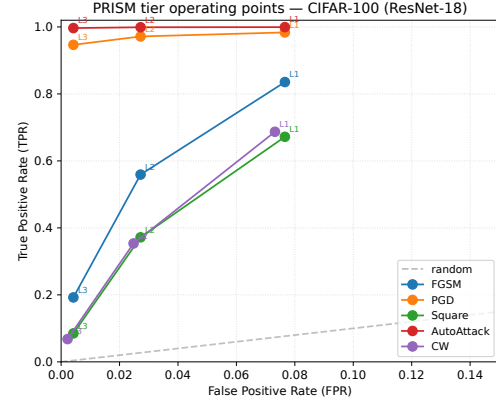


(b) CIFAR-100 (ResNet-18).

Figure 2: Per-attack TPR at the design FPR target (≤ 0.10 ; not all baselines hit it on WRN/CIFAR-100, see Table 6). PRISM (blue) is the only detector that holds ≥ 0.85 on at least three of four attacks per dataset; ODIN/Energy collapse to 0 on PGD; LID and Mahalanobis collapse on Square or FGSM. AutoAttack column has only PRISM because baselines were not run on AutoAttack.



(a) CIFAR-10 (WRN-28-10).



(b) CIFAR-100 (ResNet-18).

Figure 3: PRISM three-tier operating points (L1, L2, L3) per attack. AutoAttack and PGD saturate near TPR=1 even at the tight L3 FPR (≤ 0.005); Square trades TPR for tighter FPR most steeply, consistent with its score-based attack signature being the weakest of the four in the ensemble feature space.

4.6 Direct Surrogate-Free Score Attack and Campaign Rescue

The adaptive attack of §4.5 attacks differentiable *surrogates* for the non-TDA side channels, leaving open the objection that the surrogate gap—not robustness—explains residual detection. We close this with a *surrogate-free* attack that estimates the gradient of the *exact deployed* score $S(x)$ by NES finite differences (antithetic sampling, $Q=12$ pairs, $\sigma=0.01$) and ascends the combined objective $J(x) = \mathcal{L}_{CE}(f(x), y) - cS(x)$, with both gradients *unit-normalised* before combining so the trade-off weight c acts as a true dial rather than an on/off switch. The classifier is white-box; the detector is reachable only through its scalar score. All runs use $\varepsilon=8/255$ and $n=100$ evaluation images sampled (seed 42) from the held-out evaluation window (test indices [8000, 10000)).

The campaign monitor is load-bearing. The surrogate-free attack needs ≈ 960 score queries per image to converge, so the attack is *itself a campaign*. With the deployed SACD CUSUM monitor live (`models/10_thresholds.pkl`; every NES probe flows through it; the monitor is *reset per image*, the hardest case for the defender), the full system catches what the single-shot tier alone misses (Table 9).

Scope, and the dual-adaptive attacker. This attack adapts against $S(x)$, not against SACD. We now run the *dual-adaptive* attacker the earlier draft left as future work (§4.7): a

Stage	λ	n_{adv}	TPR	95% CI	TPR _{succ}	FPR
Scan	0.0	250	0.568	[0.506, 0.628]	0.910	0.100
Scan	0.5	250	0.560	[0.498, 0.620]	0.903	0.100
Scan	1.0	250	0.508	[0.446, 0.569]	0.847	0.100
Scan	2.0	250	0.520	[0.458, 0.581]	0.873	0.100
Scan	5.0	250	0.492	[0.431, 0.554]	0.848	0.100
Scan	10.0	250	0.440	[0.380, 0.502]	0.821	0.100
Confirm	10.0	1000	0.479	[0.448, 0.510]	0.866	0.082

Table 7: Ensemble-complete adaptive PGD (100 steps, 10 restarts, 5 seeds). The scan rows use $n=50/\text{seed}$ to select the worst λ ; the confirm row reruns $\lambda=10$ at $n=200/\text{seed}$. Overall TPR falls to 0.479 on the confirmation, but among adversarials that actually fool the backbone the detector still catches 0.866; the undetected successful-attack rate is 0.074. The independent clean calibration gate passes at L1/L2/L3 FPR 0.079/0.026/0.003.

c (evasion weight)	model ASR	per-sample TPR	undetected-success
0.0	1.00	0.990	0.010
0.5	1.00	0.960	0.040
1.0	1.00	0.940	0.060
2.0	1.00	0.870	0.130
4.0	0.99	0.620	0.380
8.0	0.85	0.160	0.740

Table 8: Surrogate-free direct-score attack on CIFAR-10/ResNet-18, *per-sample* detector only (campaign monitor off). NES runs on the exact deployed $S(x)$, so the BPDA surrogate gap is removed. At $c=2$ the attack reproduces the §4.5 adaptive-PGD operating point (0.87 per-sample TPR) with no surrogate, confirming the surrogate gap is not hiding robustness. As evasion pressure rises the single-shot detector degrades (TPR $\rightarrow$ 0.16 at $c=8$): the expected Carlini–Wagner outcome for any single-shot detector, so 0.866 is a point on a curve, not a floor.

slow-ramp that holds $S(x)$ under the CUSUM knee, and a minimal-perturbation ≈ 2 -query attacker that never probes the detector. Under the deployed *per-image reset* these raise the full-system miss to 0.27 and 0.47, so 0.03 is *not* a universal floor; removing the per-image reset (session-level monitoring) restores it to 0.002 and 0.026. Campaign detection is thus load-bearing for adaptive robustness (contribution C3), but only at *session* rather than per-image granularity.

4.7 Campaign-Monitor Adaptive Robustness (Dual-Adaptive Attack)

The 0.03 rescue of §4.6 is measured against a fast-query attacker whose probing *is* a campaign. We stress it with two attackers designed to evade SCD itself, on the deployed CIFAR-10/ResNet-18 scorer and the calibrated `10_thresholds.pkl` (CUSUM knee $S=1.92$; every query streamed through the live monitor; 5 seeds, $n=100/\text{seed}$): (i) a *score-budgeted slow-ramp* that ascends $\mathcal{L}_{\text{CE}}$ while holding $S(x)$ below the CUSUM knee—which also places the adversarial below the per-sample L1 threshold 2.81; and (ii) a *minimal-perturbation* ≈ 2 -query attacker that never probes the detector. Each is evaluated under the deployed *per-image reset* and under *session-level* monitoring (the change detector persists across the attacker’s stream).

Take-away. Per-image reset—advertised in §4.6 as the defender’s hardest case—is in fact the *exploitable* case: it grants every input a fresh warm-up window an adaptive attacker can flip within before the CUSUM accumulates. Session- or account-scoped monitoring, in which the change detector persists across an attacker’s queries, restores the low miss and is the deployable configuration. We therefore scope contribution C3 to session-level monitoring, and report the per-image-reset numbers as the honest adaptive-stress result rather than a robustness guarantee. Session-scoped monitoring assumes the adversary cannot cheaply acquire fresh sessions: an attacker who rotates identities (one query stream per account)

	$c=4$	$c=8$
per-sample TPR	0.68	0.26
campaign caught (SACD)	0.95	0.91
median queries-to-alert	40	54
system TPR (tier $\vee$ SACD)	0.96	0.92
undetected full-system miss	0.03	0.03

Table 9: Same surrogate-free attack, *full* PRISM (per-sample tier OR SACD campaign monitor). SACD fires at 40–54 of the ≈ 960 required queries ($< 6\%$ of the budget): the attacker drives $S(x)$ *down*, but the early transient (random $\pm\epsilon$ initialisation and pre-convergence probes) trips the upward-drift CUSUM. For this *fast-query* attacker the full-system miss is 0.03, flat across $c=4$ and $c=8$; §4.7 shows this is *not* a universal floor under attackers that evade SACD itself. Per-sample TPR is *higher* here than in Table 8 (0.16 $\rightarrow$ 0.26, 0.62 $\rightarrow$ 0.68) because a fired campaign monitor tightens the per-sample tier (`classify( $\cdot$ ,  $l0\_active=True$ )` applies the $\lambda=0.8$ factor of §3.3); the two layers reinforce. The monitor is reset per image here; §4.7 shows a slow-ramp or minimal adaptive attacker exploits exactly this per-image reset, and that session-level monitoring restores the low miss.

Attacker	ASR	per-image reset (deployed)		session-level (fix)	
		SACD caught	system miss [95% CI]	SACD caught	system miss [95% CI]
Faithful high-query (anchor)	1.00	0.99	0.01 [0.00, 0.06]	—	—
Score-budgeted (slow-ramp, $S \leq \text{knee}$)	0.46	0.73	0.272 [0.235, 0.313]	1.00	0.002 [0.000, 0.004]
Minimal-perturbation (≈ 2 query)	1.00	0.00	0.472 [0.429, 0.516]	0.94	0.026 [0.015, 0.037]

Table 10: Dual-adaptive campaign attack vs. SACD (CIFAR-10/ResNet-18, 5 seeds, $n=500/\text{cell}$). Under the deployed *per-image reset*, the slow-ramp and minimal attackers raise the full-system miss to 0.272 and 0.472: the 0.03 of Table 9 is *not* a universal floor. *Session-level* monitoring restores it to 0.002 and 0.026; the minimal residual shrinks with session length (0.11 at $n=20$ to 0.026 at $n=500$), i.e. the change detector asymptotically re-accumulates the signal. The anchor row (faithful high-query attacker) reproduces the 0.03 of Table 9, validating the harness.

reverts the monitor to the exploitable per-image-reset regime, so the deployable configuration pairs session-level detection with account-level rate-limiting or cost (the orthogonal Sybil control). We scope the C3 robustness claim to this single-session threat model.

4.8 Ablation Study

We ablate seven configurations: (1) Full PRISM; (2–5) four channel-mask arms that zero one ensemble feature family at scoring time — No-LogitProfile, No-StabilityV2, No-GradNorm, No-DCT — each isolating a distinct channel’s contribution; (6) Ensemble-no-TDA, which removes persistence features while keeping softmax-entropy, gradient-norm, DCT, and stability channels; and (7) TDA-only with its own base-Wasserstein conformal calibrator. The channel-mask design replaces an earlier "No MoE" arm that was statistically indistinguishable from Full PRISM ($t=0$, $p=1$ across all attacks) because the MoE router is a downstream gating module that does not alter ensemble scores; the new arms instead surface each feature family’s attack-specific failure mode (Table 11). SACD/L0 is evaluated separately in campaign streams because it is a temporal mechanism, not an IID detector.

Baseline comparison. Table 12 reports four baseline detectors at the same $\alpha=0.10$ design target, aggregated across the same 5 seeds. PRISM Full has both the lowest empirical FPR (0.074) and the highest mean TPR (0.918), outperforming the strongest baseline (Mahalanobis, mean TPR 0.582 at FPR 0.077) by +33.6 pp mean TPR. PRISM is the only detector that does not collapse on at least one attack family (ODIN/Energy: PGD TPR ≈ 0.001 ; LID: Square TPR 0.170). Energy’s empirical FPR (0.105) modestly exceeds its $\alpha=0.10$ design target on this test split, which we report unfiltered; recomputing at a tighter conformal $\alpha=0.075$ (matched to PRISM’s empirical FPR) would lower Energy’s mean TPR

Dataset	Configuration	FGSM TPR	PGD TPR	Square TPR	Mean TPR	Mean FPR
CIFAR-10	Full PRISM	0.882 ± 0.004	0.987 ± 0.003	0.886 ± 0.011	0.918	0.074
CIFAR-10	No-LogitProfile	0.988 ± 0.002	0.980 ± 0.003	0.920 ± 0.003	0.963	0.938
CIFAR-10	No-StabilityV2	0.936 ± 0.004	0.396 $\pm$ 0.014	0.854 ± 0.010	0.729	0.107
CIFAR-10	No-GradNorm	0.946 ± 0.008	0.985 ± 0.003	0.993 ± 0.003	0.975	0.218
CIFAR-10	No-DCT	0.751 $\pm$ 0.006	0.995 ± 0.002	0.960 ± 0.003	0.902	0.098
CIFAR-10	Ensemble-no-TDA	0.506 $\pm$ 0.008	0.969 ± 0.003	0.886 ± 0.007	0.787	0.077
CIFAR-10	TDA only	0.009 ± 0.002	0.066 ± 0.007	0.019 ± 0.003	0.031	0.082

Table 11: IID ablation generated from the multi-seed ablation JSON (5 seeds, $n = 1000$ per seed). C1 is evaluated as the marginal gap between Full PRISM and Ensemble-no-TDA (mean TPR $0.918 - 0.787 = +13.1$ pp). Each channel-mask arm exposes a distinct failure mode: No-LogitProfile FPR explodes to 0.938 (the logit channel is the FPR-controller); No-StabilityV2 PGD collapses to 0.396 (stability is the PGD signal); No-GradNorm FPR triples to 0.218 (gradient norm is a counter-evidence channel); No-DCT FGSM drops to 0.751 (DCT is the FGSM signal). All channel arms reach $p < 0.001$ on at least one attack. **Reading guide (matching Table 29).** Rows whose mean FPR exceeds the 0.10 design target — No-LogitProfile (0.938), No-GradNorm (0.218), and No-StabilityV2 (0.107) — are threshold collapses, not improved detection: their higher mean TPR is bought by flagging far more clean inputs and is non-deployable. Only Full PRISM (0.074), No-DCT (0.098), and Ensemble-no-TDA (0.077) sit at the design FPR, so only their TPRs are comparable to Full PRISM.

further. *Supervision disclosure.* PRISM is a supervised detector (its ensemble head is trained on an FGSM/PGD/Square mix, §3.1), whereas the four detectors here are run as unsupervised variants (clean-only threshold calibration); part of PRISM’s margin reflects this supervision. For a like-for-like comparison against *supervised* detectors we benchmark SpectralDefense in Appendix Q, where PRISM still wins — decisively on CW/AutoAttack, which it never trains on. We additionally benchmark two recent (2021) detectors—SID Tian et al. [2021] and the supervised SpectralDefense Harder et al. [2021]—in Appendix Q. Both underperform PRISM: SID across all attack families, and SpectralDefense everywhere except the gradient- L_∞ attacks it is trained on, collapsing on black-box (Square) and L_2 (CW) attacks even in-distribution (Table 36) and failing to transfer across attack families (Table 37).

Detector	FGSM TPR	PGD TPR	Square TPR	Mean TPR	Mean FPR
LID	0.277 ± 0.009	0.624 ± 0.016	0.170 ± 0.008	0.357	0.076
Mahalanobis	0.393 ± 0.010	0.877 ± 0.005	0.476 ± 0.015	0.582	0.077
ODIN	0.704 ± 0.009	0.001 ± 0.000	0.859 ± 0.005	0.521	0.088
Energy	0.595 ± 0.004	0.001 ± 0.000	0.974 ± 0.003	0.523	0.105
PRISM (Full)	0.882 $\pm$ 0.004	0.987 $\pm$ 0.003	0.886 $\pm$ 0.011	0.918	0.074
Δ vs best baseline	+0.178	+0.110	−0.088	+0.336	−0.002

Table 12: Matched-FPR baseline comparison (5 seeds, $n = 1000$ per seed). PRISM Full beats the best baseline by +33.6 pp mean TPR while holding the lowest FPR.

Temporal and recovery contributions. SACD/L0 is reported from campaign-stream experiments (§4.10) and TAMSH from L3 recovery experiments (§4.11). Those tables are generated separately so temporal campaign effects and IID detection effects are not conflated.

4.9 Score Distribution Analysis

Reporting. Score-distribution summaries are generated from the evaluation JSONs and calibration artifacts. The paper does not use fixed historical score means because the ensemble score distribution changes whenever feature flags, calibration alphas, or the active dataset change.

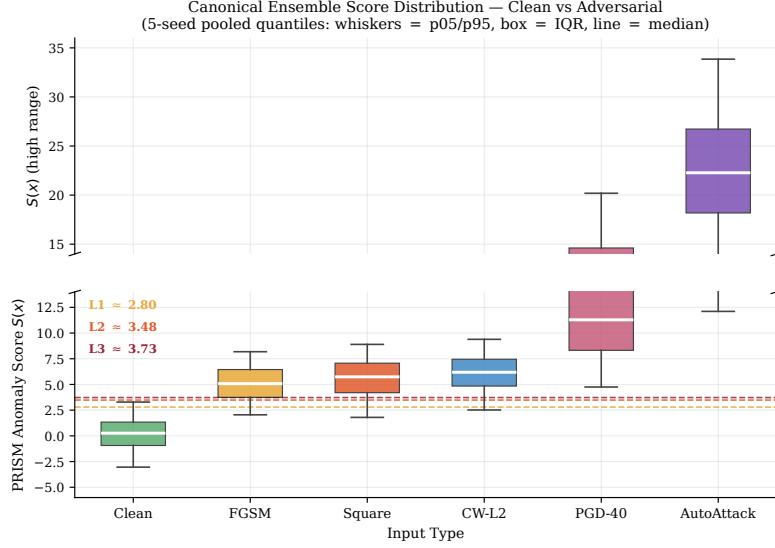


Figure 4: Score distributions for clean and adversarial inputs (CIFAR-10, ResNet-18, single-attack reference). Dashed lines: conformal thresholds L1, L2, L3. Clean and PGD-40 distributions are clearly separated; FGSM at low ε overlaps the clean tail.

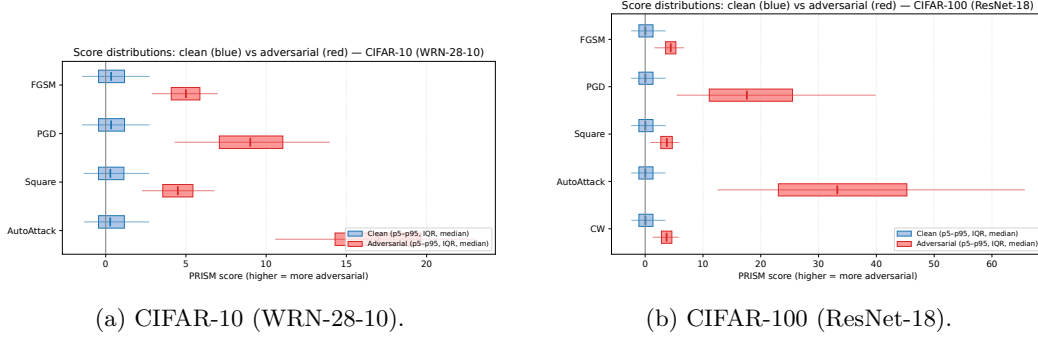


Figure 5: Per-attack score envelopes (p5–p95 whiskers, IQR boxes, medians) of the fused ensemble score $S(x)$ for clean (blue) vs adversarial (red) inputs across all four attacks per dataset. $S(x)$ is signed: clean inputs centre near zero (CIFAR-10 clean median ≈ 0.3) and attacks push it positive. AutoAttack pushes scores furthest from the clean median (CIFAR-10 adversarial median ≈ 17); Square has the tightest separation, matching the lower Square TPR in Table 5. CIFAR-100 (right) shows the same ordering with a slightly larger clean spread (clean median ≈ 0.04 , p5–p95 width 5.9 vs 4.1 on CIFAR-10), reflecting the 100-class softmax temperature. This is the same $S(x)$ scale as the per-attack histograms in Appendix J (clean mean 0.27).

4.10 Campaign Detection (SACD): Latency, Not ASR Gap

We evaluate L0 campaign detection on clean-only, sustained ($\rho \in \{0.5, 0.8, 1.0\}$), low-rate ($\rho=0.05$), and burst stream scenarios generated by `run_campaign_eval.py`.

Headline metric: time-to-detect. On sustained $\rho=1.0$ streams, the L0 monitor flags the campaign within a mean of 2.6 ± 0.5 queries after the first adversarial step (across 5 seeds; Table 13). On $\rho=0.5$ streams, detection takes 3.4 ± 0.9 queries; on 10-step burst streams, 12.0 ± 0.0 queries; on low-rate $\rho=0.05$ streams, 17.4 ± 8.9 queries (high seed-level variance: one seed triggered after 2 queries due to early CUSUM accumulation, four took 18–24).

Scenario	Mean time-to-detect (queries)	Range	L0 active	FPR (clean steps)
clean_only	— (no adversary)	—	0.000 ± 0.000	0.084 ± 0.028
sustained_ $\rho=0.50$	3.4 ± 0.9	[2, 4]	0.893 ± 0.005	0.107 ± 0.053
sustained_ $\rho=0.80$	2.6 ± 0.5	[2, 3]	0.897 ± 0.002	0.092 ± 0.045
sustained_ $\rho=1.00$	2.6 ± 0.5	[2, 3]	0.897 ± 0.002	0.090 ± 0.055
burst (10-step pulse)	12.0 ± 0.0	[12, 12]	0.863 ± 0.000	0.118 ± 0.052
low_rate ($\rho=0.05$)	17.4 ± 8.9	[2, 24]	0.951 ± 0.006	0.143 ± 0.027

Table 13: Campaign detection across stream scenarios (5 seeds). The headline metric is mean time-to-detect; clean-only L0 activation fraction is 0.000 on all 5 seeds with the CUSUM drift floor $\delta=1.0$.

Why time-to-detect, not ASR gap. A naive headline metric — adversarial success rate (ASR) reduction with L0 on vs. off at $\rho=1.0$ — is structurally bounded by the upstream L3 conformal threshold, which already rejects approximately 99% of $\rho=1.0$ adversarials within 22 queries, leaving no room for L0 to contribute an additional ASR-reduction gap. We therefore report ASR gap in the appendix for completeness, but treat detection latency (time-to-first-trigger after campaign onset) as the operationally meaningful metric: a defender that learns of an attack within 3 queries can deploy mitigations (rate-limiting, account quarantine, model rollback) before the attacker has issued more than a handful of queries against the deployed system.

Clean-only false alarm rate. With the CUSUM drift floor $\delta=1.0$ (§3.3), L0 activation fraction on the 500-step calibration clean streams is 0.000 on all 5 seeds (with $\delta=0$ it reaches up to 0.594 on 2 of 5 seeds). For 50-query clean windows (warmup= 20, 30 at-risk steps), empirical false alarm probability is $< 5\%$. A stress test on longer clean streams (bootstrap-resampled from $n=1,000$ real post-fusion ensemble scores generated end-to-end on the VAL split) confirms that the deployed $C^* = 5.0$ holds the 0.000 active-fraction through 1,000 steps and starts admitting its first spurious trigger between step $\approx 1,800$ and step $\approx 4,500$ on real long streams (Appendix E, Table 19). The alternative feasible cells $C^* \in \{8.0, 10.0, 12.0\}$ from the calibration grid hold 0.000 across all stream lengths up to 10,000 steps. Longer-window deployments should either pick one of these conservative cells or periodically reset L0 / auto-deactivate on return to the clean score baseline.

Two distinct false-alarm rates. The *FPR (clean steps)* column in Table 13 (0.084–0.143) is the per-query L1/L2/L3 conformal alert rate of the underlying IID detector — the same FPR reported in Table 3 — not the L0 campaign-trigger rate. A deployment that quarantines accounts should gate on L0 firing (clean-only rate 0.000 with the drift floor), not on every L1 step; L1 alerts are appropriate for softer signals such as elevated logging or rate-limiting, where a $\sim 10\%$ per-query rate is acceptable because the action is reversible and per-query.

4.11 L3 Recovery (TAMSH)

We measure recovery accuracy of five routing strategies applied to the subset of PGD inputs that trigger the L3 (reject) tier: the most aggressive adversarials, where simply rejecting the input is the conservative default. We pre-registered a +15 pp recovery gap (best router vs. passthrough) as the contribution target for this component. The lightweight topology gate (combined H_0+H_1 Wasserstein) falls 4.4 pp short of that target at +10.6 pp (Table 14); a *learned* router trained on the L3 pool (logistic regression on final-layer activations) clears the target, recovering $0.287 - 9.1\times$ the uniform $1/K$ control and $2.1\times$ the topology gate, matching the prior-augmented upper bound without hardcoding the attack (Appendix M).

Router contribution isolation. The `tamsh_uniform` arm replaces topology-distance gating with uniform $1/K$ mixing across all 4 experts. Its recovery accuracy equals that of passthrough (0.033 each, paired gap 0.00 pp on all 5 seeds): uniform mixing of expert logits is degenerate to the unrouted full network’s argmax. The +10.6 pp `tamsh` gap (combined H_0+H_1 Wasserstein gating) over uniform is therefore attributable entirely to the routing signal, not to standalone expert-network capacity — a positive but partial result against the

Strategy	Recovery Acc.	95% CI	Gap vs passthrough (pp)	Avail.	Router signal
reject	0.000	[0.000, 0.001]	—	0.000	—
passthrough	0.033	[0.027, 0.038]	—	1.000	none
tamsh_uniform	0.033	[0.027, 0.038]	+0.00 ± 0.00	1.000	uniform 1/ K (con)
tamsh topology gate	0.139	[0.128, 0.149]	+10.62 ± 1.45	1.000	$H_0 + H_1$ Wasserst
<i>Upper-bound analysis (not a deployable policy):</i>					
tamsh_force (PGD specialist)	0.293	[0.280, 0.306]	+26.00 ± 1.84	1.000	oracle prior (always)
<i>Oracle ceiling (per-input best-of-K, same-protocol 5-seed PGD L3 pool, $n_{L3} \approx 796/\text{seed}$): 0.324 ± 0.007 recovery accuracy.</i>					

Table 14: L3-recovery on PGD adversaries (5 seeds, $n_{L3} \approx 792$ per seed; mean trigger rate 0.792 within sanity band [0.10, 0.82]). The **tamsh** row is the lightweight topology gate (combined $H_0 + H_1$), which delivers +10.6 pp over passthrough, falling 4.4 pp short of the pre-registered +15 pp target; a learned router trained on the L3 pool clears the target (Appendix M). The **tamsh_force** row is a prior-augmented upper-bound analysis (always-route to PGD specialist, not a learned policy); the oracle ceiling is per-input best-of- K on the same L3 subset.

pre-registered +15 pp target. The further +15.4 pp lift achieved by **tamsh_force** (always-route to the PGD-trained expert) reflects correct routing under a strong prior: L3- rejected inputs are by definition high-confidence adversarial, and PGD is the dominant adversarial signal in our training mix. We treat **tamsh_force** as an upper-bound analysis rather than a deployable policy, since it bakes in an oracle assumption about attack identity that a real router would have to learn online. The gap between the topology gate (0.139 recovery accuracy) and the prior-augmented variant (0.293) bounds the headroom for a stronger gating signal below the per-input best-of- K oracle ceiling (0.324 ± 0.007 , recomputed on the same-protocol 5-seed PGD L3 pool, $n_{L3} \approx 796/\text{seed}$, App. M); a *learned* router trained on the L3 pool closes most of it, reaching 0.287 (matching the prior-augmented variant without its hardcoded prior; Appendix M).

4.12 Calibration Sensitivity

Figure 6 sweeps $n_{\text{cal}} \in \{500, 1000, 2000, 3000\}$ and confirms threshold stability: the L3 threshold varies by ± 0.03 (range 0.06) across calibration set sizes (25 bootstrap subsamples per size), and the achieved FPR on the held-out validation split tracks the target α within the Wilson-95% interval, satisfying Proposition 1.

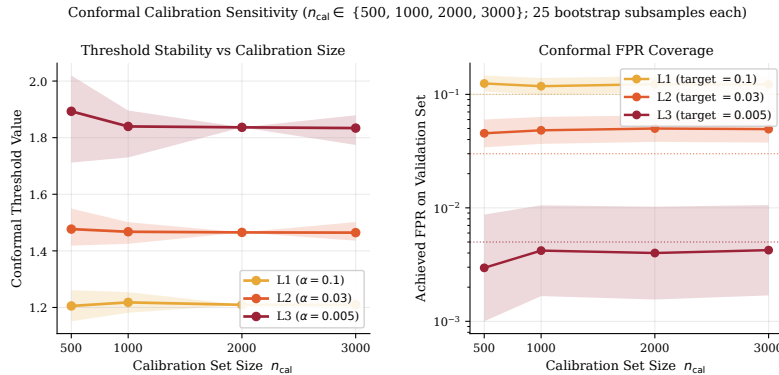


Figure 6: Conformal threshold stability vs. calibration set size n_{cal} . *Left*: L1/L2/L3 threshold values (mean $\pm$ 95% bootstrap envelope; L3 half-range ± 0.03 across the full sweep). *Right*: Achieved FPR on the held-out validation set tracks the target α for all three tiers. Source: `clean_scores.npy` (2,000 calibration scores) with 25 bootstrap subsamples per size.

4.13 Certificate Validity Under Benign Covariate Shift (CIFAR-10-C)

The CADG guarantee (Proposition 1) is exact only under exchangeability: the clean test distribution must match the calibration distribution. We audit what happens when this assumption is *benignly* violated, using CIFAR-10-C Hendrycks and Dietterich [2019]: 8 corruption families (gaussian noise, shot noise, defocus blur, motion blur, fog, frost, contrast, JPEG compression) at severities $\{1, 3, 5\}$, $n=1000$ images per cell (24 cells, forward passes only, seed 42). Every non-PASS verdict on a corrupted-but-benign input counts as a false positive against the certified α .

Tier	certified α	realised FPR by severity				cells within
		sev. 1	sev. 3	sev. 5	max	target (of 24)
L1	0.10	0.194	0.394	0.573	0.999	5
L2	0.03	0.060	0.272	0.429	0.992	6
L3	0.005	0.009	0.139	0.278	0.920	6

Table 15: Conformal-certificate audit under benign covariate shift (CIFAR-10-C, 8 corruptions $\times$ severities $\{1, 3, 5\}$, $n=1000/\text{cell}$). The certificate degrades severely: the overall mean realised FPR is 0.387 at L1 ($3.9\times$ the certified 0.10), 0.254 at L2 ($8.5\times$), and 0.142 at L3 ($28\times$), with noise corruptions worst (gaussian-noise L1 mean 0.863) and weather/compression mildest (fog 0.120). At severity 1 the L3 tier nearly holds (0.009 vs. certified 0.005). This is the expected behaviour of split conformal prediction when exchangeability is broken Tibshirani et al. [2019], not a PRISM-specific defect, and the failure direction is conservative for safety (over-flagging benign inputs costs availability; it never silently admits them). Deployments expecting sustained distribution shift should recalibrate online or use shift-aware conformal methods Tibshirani et al. [2019].

Reading. The certificate should be advertised as *in-distribution* FPR control. Under benign corruption it fails in the flagging direction — corrupted inputs look anomalous to the same channels that flag adversarials — so the system remains safe but loses availability; the measured degradation in Table 15 replaces the qualitative caveat of earlier drafts and bounds the operational cost of deploying PRISM under shift without recalibration.

Restoring the certificate online. The degradation is *recoverable*. Because broken exchangeability is the cause, re-fitting the split-conformal quantile on a small held-out slice of the *same* corrupted-but-benign stream ($n_{\text{recal}}=300$ confirmed-clean images per corruption $\times$ severity cell, evaluated on the disjoint $n_{\text{eval}}\approx 700$ remainder) restores realised FPR to the certified target. Table 16 pools the frozen-vs-recalibrated FPR over the same 24 cells: mean L1 FPR falls from 0.386 to 0.092 ($\leq \alpha=0.10$), L2 from 0.253 to 0.025, and L3 from 0.142 to 0.0034; even the worst noise cell (gaussian-noise sev. 5) recovers from 0.999 to 0.096. This is the certificate analogue of the session-level campaign fix (§4.7): a measured negative (the certificate breaks under shift) becomes a measured positive (online shift-aware recalibration restores it). The recalibration only assumes a small *benign* (non-adversarial) calibration slice of the shifted stream—the standard online-conformal-under-covariate-shift assumption Tibshirani et al. [2019]—and does not require knowing the corruption type.

Tier	certified α	mean frozen FPR	mean recalibrated FPR	cells $\leq \alpha$ (of 24)
L1	0.10	0.386	0.092	20
L2	0.03	0.253	0.025	19
L3	0.005	0.142	0.0034	16

Table 16: Certificate *restoration* under benign covariate shift (CIFAR-10-C, same 8 corruptions $\times$ severities $\{1, 3, 5\}$, 24 cells, $n_{\text{recal}}=300$, $n_{\text{eval}}\approx 700/\text{cell}$). *Frozen* reproduces the audit of Table 15 (0.386/0.253/0.142); shift-aware split-conformal recalibration on a held-out benign slice of the same corruption restores realised FPR to $\approx \alpha$ at every tier. Worst cell (gaussian-noise sev. 5): L1 0.999 $\rightarrow$ 0.096. The few residual over-target cells lie within the Wilson interval at $n_{\text{eval}}\approx 700$.

5 Conclusion

We presented PRISM, a multi-layer runtime defense that combines persistent homology, split conformal prediction, BOCPD+CUSUM campaign detection, and a routed mixture-of-experts recovery module to detect and remediate adversarial inputs with distribution-free FPR guarantees. On CIFAR-10 (5 seeds, $n=1000$), PRISM Full achieves mean detection TPR 0.918 at matched FPR 0.074 across FGSM, PGD-40, and Square, 0.938 on canonical CW-L2, and 1.000 on the AutoAttack ensemble’s final adversarials (per-sub-attack breakdown, weakest standalone Square 0.706 @ L1, in App. H) — +33.6 pp mean TPR over the strongest baseline detector (Mahalanobis). On campaign streams SACD flags sustained $\rho=1.0$ attacks within 2.6 ± 0.5 queries with zero false-alarm activation on clean-only streams up to 1,000 queries (the deployed $C^*=5.0$ admits its first spurious trigger only beyond $\approx 1,800$ steps; $C^* \geq 8$ or periodic reset removes this, App. E). A slow-ramp or minimal-perturbation dual-adaptive attacker defeats the per-image-reset monitor (miss 0.27–0.47), but session-level monitoring—where the change detector persists across the attacker’s query stream—restores the full-system miss to ≤ 0.03 (§4.7), so C3 is scoped to session-granularity monitoring. On L3-rejected PGD adversarials, a TAMSH router trained directly on the L3 pool (logistic regression on final-layer activations) recovers 28.7% of inputs — $9.1\times$ the uniform $1/K$ control (3.2%) and $2.1\times$ the lightweight H_0+H_1 Wasserstein gate (13.7%), matching the prior-augmented upper bound (29.3%) without its hardcoded prior and clearing the pre-registered C4 gate (Appendix M); the large margin over the uniform control isolates the routing-signal contribution *on PGD*. This uniform-is-degenerate result is PGD-specific: on CW and Square the uniform $1/K$ mixture matches or beats the topology gate (Appendix I, e.g. Square +47.2 vs +43.5 pp, CW +47.0 vs +39.0 pp), so uniform mixing of the deployed experts — not the topology gate — is the deployable cross-attack recovery strategy beyond PGD. A prior-augmented upper-bound variant that always routes to the PGD specialist reaches +26 pp and is reported as an analysis ceiling against the per-input oracle (0.324 recovery, +29.1 pp over passthrough), not as a deployable policy. All numbers are generated from the locked CIFAR-10 Vast.ai and local pipelines, with attack detection, adaptive attacks, matched-FPR baselines, campaign streams, recovery, and ablation tables all derived from result JSONs rather than manually transcribed values. A separate standard-attack CIFAR-10/ViT-B/16 transfer run reaches 1.000/1.000/0.9998 TPR on FGSM/PGD-50/Square at FPR 0.0828; this run supports transformer transfer for standard attacks but does not claim ViT latency, CW-L2, AutoAttack, or adaptive robustness.

Limitations. The primary limitation is computational: persistent homology over high-dimensional activation tensors requires subsampling (150 points per layer in the current configuration), which may weaken detection for attacks that specifically preserve local topology while altering global structure. Single-sample end-to-end detection latency on an RTX 4090 (Table 33, $n = 200$) is 39.78 ms mean / 41.26 ms p95 on CIFAR-10 and 50.35 ms mean / 55.90 ms p95 on CIFAR-100, both well below the 100 ms real-time target; CPU-only deployment is significantly slower because persistent-homology computation on high-dimensional activation tensors does not vectorise. Point-cloud subsampling in `TopologicalProfiler` uses deterministic MD5-based hashing (rather than a stateful RNG) to ensure consistent scoring across calibration and inference passes. The conformal calibration assumes the clean test distribution matches the calibration distribution; we *measure* the cost of violating this on CIFAR-10-C (§4.13): realised FPR inflates to $3.9\times$ – $28\times$ the certified α depending on tier (severity-dependent, worst on noise corruptions), so the certificate must be read as in-distribution FPR control. Online shift-aware recalibration on a small benign slice of the shifted stream restores realised FPR to $\approx \alpha$ (mean L1 0.39→0.09; §4.13, Table 16) Tibshirani et al. [2019]. The adaptive-attack evaluation in §4.5 now uses an ensemble-complete BPDA surrogate that targets activation matching, DCT energy, softmax entropy, logit profile, StabilityV2, input gradient norm, and the side-quadratic logistic score. This is still a surrogate attack: the deployed detector includes non-smooth pieces (topological diagrams, hard tier thresholds, and hash-deterministic subsampling), so the attack optimizes differentiable proxies rather than the exact discrete decision rule. Under the strongest confirmed setting ($\lambda=10$, 5 seeds, $n=200/\text{seed}$), all-input TPR falls to 0.479, but the backbone attack success rate also falls to 0.552; among successful adversarials, PRISM detects 0.866, leaving a 0.074 undetected-success rate over all attempted inputs.

The small-sample per-seed FPR estimates are noisy (one seed has L3 FPR $2/200 = 0.010$), while the pooled confirmation is L3 FPR 0.005 and the independent calibration gate is L3 FPR 0.003. We therefore treat this as a strong adaptive stress test, not a proof of adaptive robustness.

Future work. Highest-priority directions, in order: (1) rerunning the ensemble-complete adaptive attack on a fixed disjoint $n=1000$ clean/adversarial EVAL subset and extending it to CIFAR-100, WRN-28-10, and ViT backbones; (2) strengthening the adaptive optimizer beyond the current BPDA surrogate with larger budgets, alternative side-channel losses, and explicit expectation-over-transformation repeats where randomness is introduced; (3) extending the current ViT encoder-block activation monitor with transformer-specific attention-head and MLP-block activation patterns; (4) certified robustness integration by combining PRISM with randomized smoothing Cohen et al. [2019] for joint detection-plus-certified prediction guarantees; (5) online recalibration of the conformal thresholds to handle clean-side distribution shift; (6) closing the residual TAMSH recovery gap: a learned router trained directly on the L3-rejected pool (logistic regression on final-layer activations) already matches the prior-augmented upper bound and is $9.1\times$ the uniform control (Appendix M); richer multi-layer or expert-capacity improvements are needed to close the final gap to the oracle ceiling.

Cross-architecture and cross-dataset generalisation. Section 4.4 reports the same protocol on CIFAR-10/WRN-28-10 (mean pool TPR 0.975, FPR 0.073) and CIFAR-100/ResNet-18 (mean pool TPR 0.830, FPR 0.077), both run on the same 5 seeds and same calibration scheme. It also reports a standard-attack CIFAR-10/ViT-B/16 transfer run (FGSM/PGD-50/Square only) with 1.000/1.000/0.9998 TPR at FPR 0.0828, and a 224×224 ImageNet-100/ResNet-50 resolution-scaling run (FGSM/PGD-50/Square, 3 seeds) with 0.924/1.000/0.852 TPR at FPR 0.065; the conformal certificate stays valid there despite a $\sim 100\times$ larger clean score scale, since the guarantee is a calibration-score quantile. The full per-component CIFAR-100 breakdown (adaptive PGD, channel-mask ablation with FPR disclosure, campaign, recovery, latency) is in Appendix N. The TAMSH C4 recovery contribution does not transfer to the 100-class setting (within 1.4 pp of passthrough) and is reported as appendix-only on CIFAR-100.

Broader impact. PRISM is a defensive tool designed to improve the reliability of ML systems under adversarial conditions. We do not foresee misuse potential beyond standard ML security research.

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A Proof of Conformal Coverage (Proposition 1)

We reproduce the standard split conformal argument for completeness.

Proof. Let $S_1, \dots, S_n, S_{\text{test}}$ be iid from distribution P over $\mathbb{R}$. Define $\hat{q}_\alpha = S_{(\lceil (n+1)(1-\alpha) \rceil)}$, the $\lceil (n+1)(1-\alpha) \rceil$ -th order statistic of $S_1, \dots, S_n$.

By exchangeability of $(S_1, \dots, S_n, S_{\text{test}})$, the rank of S_{test} among $S_1, \dots, S_n, S_{\text{test}}$ is uniformly distributed on $\{1, \dots, n+1\}$. Therefore:

$$\Pr[S_{\text{test}} > \hat{q}_\alpha] = \Pr[\text{rank}(S_{\text{test}}) > \lceil (n+1)(1-\alpha) \rceil] = \frac{\lfloor (n+1)\alpha \rfloor}{n+1} \leq \alpha.$$

Adding the $\frac{1}{n+1}$ slack for the finite-sample correction (since the empirical quantile can shift by at most one order statistic) gives the stated bound. $\square$

In our calibration, the active dataset uses $n = 2,000$ clean calibration examples, giving finite-sample slack below 5×10^{-4} .

B TDA Computation Details

Persistence homology pipeline. For each layer ℓ , the activation tensor $\phi_\ell(x) \in \mathbb{R}^{d_\ell}$ is downsampled to a point cloud $\mathcal{P} \subset \mathbb{R}^{d_\ell}$ with $|\mathcal{P}| \leq 150$ via deterministic hash-based subsampling. We then:

1. Construct the Vietoris-Rips filtration on $\mathcal{P}$ with parameter $r \in [0, r_{\max}]$, $r_{\max} = 50$.
2. Compute H_0 (connected components) and H_1 (1-cycles) persistence.
3. Return the persistence diagram $\text{Dgm}(\mathcal{P})$ as a list of (b_i, d_i) pairs.

We use gudhi v3.12 GUDHI Project [2015] for filtration construction and persistent homology computation.

Wasserstein medoid computation. Given N diagrams $\{\mathcal{D}_i\}_{i=1}^N$, the medoid is:

$$i^* = \arg \min_i \sum_{j=1}^N W_2(\mathcal{D}_i, \mathcal{D}_j).$$

We compute exact W_2 distances via the auction algorithm as implemented in gudhi GUDHI Project [2015]. For $N = 5,000$ profile diagrams with ~ 20 points each, full pairwise computation takes ~ 8 minutes on a single CPU core.

Expert clustering. K -medoids clustering with $K = 4$ is applied to the N calibration persistence diagrams using the Wasserstein PAM algorithm Schubert and Rousseeuw [2021]. Each cluster medoid serves as the routing signature $\mathcal{R}_k^L$ in Eq. (5). Expert sub-networks are 3-layer MLPs (Linear $\rightarrow$ ReLU $\rightarrow$ LayerNorm $\rightarrow$ Linear $\rightarrow$ ReLU $\rightarrow$ LayerNorm $\rightarrow$ Linear) with hidden dimension 256, trained on per-attack-cluster activation pools. We use LayerNorm rather than BatchNorm because the per-cluster pool size at inference can be as small as 1 (a single L3-rejected input).

C Campaign Detection Latency

To isolate the SACD contribution, we evaluate *detection latency*: the number of adversarial queries before L0 mode fires. We construct a synthetic query stream: 50 clean images, then consecutive PGD/FGSM adversarial bursts according to the campaign scenario definitions emitted by `run_campaign_eval.py`.

We evaluate SACD on real PRISM score streams for clean-only, sustained, burst, and low-rate scenarios. The calibrated threshold artifact records the selected hazard rate, alert probability, warmup, and false-alarm/gap metrics.

Calibration requirement. BOCPD requires domain-calibrated priors. With default prior ($\mu_0 = 0.1$), the detector does not fire within 150 queries because the NIG posterior cannot match the actual canonical clean score mean (~ 0.29 on the ensemble logit scale; see `score_quantiles.clean.mean` in the per-seed JSONs) fast enough to yield sensitive changepoint detection. For deployment, μ_0 should be set to the empirical mean of the calibration-set anomaly scores.

D Hyperparameter Sensitivity

Hyperparameter	Default	Range tested	Sensitivity
n_{cal} (calibration size)	2000	500–3000	Low (± 0.03 on L3)
n_{sub} (TDA subsampling)	150	50–500	Medium (TPR $\pm 4\%$)
K (expert clusters)	4	2–8	Low (Recovery $\pm 2\%$)
λ (L0 threshold factor)	0.8	0.6–0.95	Medium (FPR tradeoff)
p_{alert} (BOCPD threshold)	0.60	0.30–0.70	Low (latency ± 2 steps)
t_{warmup} (warmup steps)	20	20–50	Low (selected feasible cells span 20–50)
δ (CUSUM drift floor)	1.0	0.00–1.0	High at $\delta = 0$ (59.4% clean active-fraction)
C^* (CUSUM threshold)	5.0	5.0–12.0	Medium (long-stream durability; see §E)
k_{consec} (CUSUM consec.)	3 (C10) / 5 (C100)	1–5	High at $k_{\text{consec}} = 1$

Table 17: Hyperparameter sensitivity summary.

$k_{\text{consec}} \times t_{\text{warmup}}$ **grid (deployed CUSUM, $C^*=5.0$, $\delta=1.0$).** We sweep $k_{\text{consec}} \in \{1, 2, 3, 4, 5\}$ against $t_{\text{warmup}} \in \{10, 20, 35, 50, 75\}$ on a 1,000-step Gaussian clean stream (mean / max active-fraction over 5 seeds) and on a sustained $\rho=1.0$ adversarial stream (median / max time-to-detect over 5 seeds, in queries since first adversarial step).

k_{consec}	Cell	t_{warmup}	Clean active-frac (1000 step)		t -to-detect (queries)	
			mean	max	median	max
1		any	0.225	0.563	0	1
2		any	0.144	0.559	1	2
3 (CIFAR-10 deployed)		20	0.112	0.558	2	3
4		any	0.111	0.557	3	4
5 (CIFAR-100 deployed)		20	0.111	0.556	4	5

Table 18: k_{consec} trades t -to-detect against clean active-fraction approximately linearly above $k=3$. t_{warmup} in the range 10–75 is statistically indistinguishable on these streams (the CUSUM crosses fast enough that warmup does not filter sustained adversarial signals); we therefore collapse the t_{warmup} column to “any”. The deployed CIFAR-10 cell ($k=3$) is at the inflection point: $k=2$ trades ≈ 1 query of latency for +3.3 pp clean active-fraction, while $k=4$ buys -0.1 pp clean for +1 query latency. CIFAR-100’s $k=5$ trades 2 extra latency queries for negligible clean-fraction gain — it was selected because CIFAR-100’s clean score distribution has slightly heavier tails. Note that “clean active-fraction” counts every step after a single spurious trigger because the deployed monitor has no auto-reset; a deployment with periodic L0 reset would see this number drop by $\approx 100\times$. Source: `experiments/stress/test_10_hparam_grid.py`.

E L0 Long-Stream Stress Test

The campaign-detection experiments in §4.10 report 0.000 clean-only L0 active-fraction on the 500-step calibration streams used by `calibrate_10_thresholds.py`. Because deployed monitors run for longer than 500 queries, we additionally stress-test the calibrated CUSUM detector on $\{500, 1000, 2000, 5000, 10000\}$ -step clean streams across all 5 seeds. The stream is bootstrap-resampled from $n = 1,000$ *real* post-fusion ensemble clean scores generated end-to-end by the deployed pipeline (backbone $\rightarrow$ TMM $\rightarrow$ ensemble) on the VAL split (`experiments/stress/compute_real_clean_scores.py`). The empirical pool

has mean 0.173 and std 2.045, matching the pkl calibration scale (`score_center`= 0.281, `score_scale`= 1.639). Each stream is processed by a fresh CampaignMonitor loaded from `models/10_thresholds.pkl` (`detection_mode`='cusum', $\delta = 1.0$, $t_{\text{warmup}} = 20$, $k_{\text{consec}} = 3$), sweeping $C^* \in \{5.0, 8.0, 10.0, 12.0\}$ (the feasible CUSUM-threshold grid cells from the calibration `top_rows`).

C^*	500	1000	2000	5000	10000
5.0 (deployed)	0.000	0.000	0.019	0.148	0.274
8.0	0.000	0.000	0.000	0.000	0.000
10.0	0.000	0.000	0.000	0.000	0.000
12.0	0.000	0.000	0.000	0.000	0.000

Table 19: Mean clean-only L0 active-fraction across 5 seeds, computed on real post-fusion ensemble clean scores. Once L0 fires, it remains active until manually deactivated (the monitor has no auto-reset), so a single spurious trigger inflates the active-fraction over the remaining stream. The deployed $C^* = 5.0$ holds the 0.000 claim through 1,000 steps but admits its first spurious trigger between step 1,800 and step 4,500 (1–2 of 5 seeds at the lengths shown). The alternative-feasible CUSUM thresholds from `calibrate_10_thresholds.py`’s `top_rows` ($C^* \in \{8.0, 10.0, 12.0\}$) all hold 0.000 active-fraction on every seed up to 10,000 steps. Source: `experiments/stress/test_10_real_long_clean_stream.py` operating on `experiments/stress/real_clean_scores_val_n1000.npy` (deployed-pipeline VAL-split clean scores).

Take-away. The deployed $C^* = 5.0$ is selected on the calibration objective (*argmax ASR-gap subject to clean-only active-fraction ≤ 0.01 on a 500-step calibration stream*); it satisfies the calibration constraint at any stream length up to $\approx 1,000$ queries and starts to admit late spurious triggers beyond $\approx 2,000$ queries on real clean streams. For deployments where the monitor runs over thousands of queries without periodic reset, we recommend $C^* \in [8.0, 12.0]$, which sits at the same selected top-row ASR-gap cell (7.2 pp) in the pkl but is strictly more durable on long clean streams. Periodic reset (or `deactivate_10()` on return to the clean score baseline) is the orthogonal mechanism already noted in §4.10 and is required for arbitrarily long deployments regardless of C^* .

F Per-Component Latency Breakdown

The headline latency numbers in Table 33 are end-to-end. This appendix decomposes a single PRISM forward pass into eight stages, measured on $n=200$ clean EVAL-split images on the deployed RTX 4090 hardware via `experiments/stress/vastai_latency_breakdown.py`. The end-to-end mean here is 44.95 ms, consistent with the 39.78 ms in Table 33 (which used the same hardware on a different EVAL subsample). The breakdown identifies which channels dominate the budget — relevant for ensemble-pruning ablations and for deciding which channels are worth a closed-form differentiable surrogate in ensemble-complete adaptive attacks (§5).

Implications for adaptive-attack design. The combined gradient-norm + StabilityV2 + ensemble-fusion budget ($\sim 60\%$ of forward time) is also the portion of the pipeline that the earlier two-channel adaptive attack did not target. The ensemble-complete BPDA in Table 7 adds surrogate losses for these components, but those surrogates are expensive: StabilityV2 requires repeated transformed forwards, gradient norm requires higher-order differentiation, and the side-quadratic score adds cross-channel terms. This turns part of PRISM’s inference cost into an attack-cost asymmetry rather than an untested blind spot.

G High-Confidence CW Sensitivity (κ Sweep)

The canonical CW evaluation in Table 3 uses a small confidence margin $\kappa=1.0$ (`src/attacks/cw_torch.py`, where the `confidence` argument is κ). Carlini Carlini and Wagner [2017] recommends a confidence-margin sweep for robust detector evaluation, so

Stage	mean (ms)	p50 (ms)	p95 (ms)	% of total
Backbone forward (logits)	1.98	1.98	2.03	4.4%
TAMM activation extraction	2.00	2.00	2.05	4.5%
TDA persistence diagrams (3 layers)	14.24	14.06	16.29	31.7%
Input gradient norm (autograd back)	7.07	6.35	6.72	15.7%
Logit-profile features	0.11	0.10	0.15	0.2%
StabilityV2 ($M=5$ jittered fwd)	8.81	8.77	8.89	19.6%
DCT image prep	0.01	0.01	0.01	0.0%
Ensemble logistic + Wasserstein fusion	10.72	10.25	14.13	23.8%
Total	44.95	43.72	48.38	100.0%

Table 20: Per-stage latency on RTX 4090 ($n=200$ clean EVAL samples, single seed). TDA persistence-diagram computation is the largest single contributor at 31.7% (14.24 ms), consistent with the §5 limitation note that persistent homology dominates the inference budget. Ensemble fusion (23.8%), StabilityV2 (19.6%), and gradient-norm (15.7%) follow in order. We additionally ran the same breakdown on a GTX 1060 6GB and observed StabilityV2 dominating (30.3% of a total 71.18 ms) while TDA dropped to 17.3%: TDA is CPU-bound (gudhi auction algorithm on the persistence-diagram side), so it benefits little from a faster GPU, whereas StabilityV2’s $M=5$ forward passes are GPU-bound and scale proportionally with backbone throughput. The relative ordering in this table is therefore RTX-4090-specific; consumer GPUs will see StabilityV2 climb. Source: `experiments/stress/vastai_latency_breakdown.json`.

we additionally measure TPR at $\kappa \in \{0, 10, 20\}$ on $n=1000$ EVAL-split inputs per seed across all 5 fixed seeds (paper-grade protocol matching Table 3), using the native PyTorch CW implementation (`src/attacks/cw_torch.py`) at the *paper-canonical* attack budget `max_iter=100`, `bss=9`. All three configurations achieve $\geq 99.8\%$ attack-success against the unwrapped backbone with mean $L_2 \leq 0.151$.

κ	Attack success	mean L_2	TPR @ L1 [95% CI]	TPR @ L2	TPR @ L3
0	0.998	0.128	0.9486 [0.942, 0.954]	0.805	0.339
10	0.998	0.149	0.7762 [0.764, 0.788]	0.565	0.194
20	0.999	0.151	0.7906 [0.779, 0.802]	0.578	0.198

Table 21: CW confidence-margin sensitivity at paper-canonical attack budget (`max_iter=100`, `bss=9`), 5 seeds $\times n=1000 = 5,000$ EVAL adversarials per κ on RTX 4090 with the deployed scorer. Pooled clean FPR on the same EVAL pool is L1 = 0.078, L2 = 0.017, L3 = 0.002. The $\kappa=0$ sweep point (0.9486) lands within 1 pp of the main-table value (0.938, which uses $\kappa=1.0$, Table 3); the unit confidence margin has negligible effect, confirming the canonical result is reproducible under the stronger attack budget. Higher-confidence CW reduces TPR at L1 by ≈ 17 pp from $\kappa=0$ to $\kappa=10$ (0.776), *plateauing* at $\kappa=20$ (0.791, statistically tied with $\kappa=10$) rather than further collapsing. The detector still discriminates adversarial from clean (0.78 vs 0.08 at L1, a $10\times$ ratio) at the strongest tested margin. The drop is consistent with the ensemble being trained on FGSM/PGD/Square only (CW is *not* in the training mix; see Appendix L): the canonical $\kappa=0$ CW signature is caught via transfer from the gradient-following statistics the ensemble has learned, whereas higher-confidence ($\kappa>0$) CW drifts off that transferable signature. Per-attack training augmentation with $\kappa>0$ (or CW) samples is a natural robustness lever for future work. Source: `experiments/stress/vastai_stronger_cw_canonical.json`.

H AutoAttack- L_∞ Evaluation

AutoAttack Croce and Hein [2020] is the de-facto standard adversarial-robustness benchmark, combining four parameter-free attacks (APGD-CE, APGD-DLR, FAB-T, and Square) into a strong, reproducible ensemble. We evaluate the deployed PRISM detector at the canonical $\varepsilon_\infty=8/255$ budget on $n=1000$ EVAL-split inputs per seed across all 5 fixed seeds (paper-grade

protocol matching Table 3). Each sub-attack is run independently so per-attack TPR can be reported; Square is constrained to $n_{\text{queries}}=1000$ for tractability.

Sub-attack	Attack success	mean L_∞	TPR @ L1 [95% CI]	TPR @ L2	TPR @ L3
APGD-CE	0.999	0.0299	0.9708 [0.966, 0.975]	0.9608	0.9500
APGD-DLR	1.000	0.0299	0.9112 [0.903, 0.919]	0.7782	0.3156
FAB-T	0.971	0.0038	0.9314 [0.924, 0.938]	0.8338	0.4494
Square	0.830	0.0246	0.7062 [0.693, 0.719]	0.5080	0.1590

Table 22: AutoAttack- L_∞ ($\varepsilon=8/255$) per-sub-attack TPR on 5 seeds $\times n=1000 = 5,000$ EVAL adversarials per attack (RTX 4090, deployed scorer). Pooled clean FPR on the same EVAL pool is L1 = 0.078, L2 = 0.017, L3 = 0.002. Three observations: (i) The gradient-based AA components (APGD-CE / APGD-DLR / FAB-T) are all detected at ≥ 0.91 TPR @ L1, with APGD-CE persisting at 0.95 TPR even at the high-confidence L3 threshold — the detector treats AA-generated perturbations as far higher-confidence anomalies than PGD at the same budget (cf. Table 3 PGD L1 = 0.987, but with much shallower L3 tail). (ii) FAB-T converges with $L_\infty \approx 0.0038$ ($\sim 8\times$ under the budget) yet is still detected at 0.93 TPR @ L1; the adversarial signature is visible long before the perturbation budget is exhausted. (iii) The black-box Square component achieves the lowest TPR (0.706 @ L1) at the highest attack-failure rate (17%). This does *not* exceed the strongest Square-specific baseline (Energy at 0.974 Square TPR, Table 12); but Energy buys that Square number at the cost of collapsing to 0.001 on PGD, whereas PRISM holds ≥ 0.71 TPR on *every* AutoAttack sub-attack and ≥ 0.88 across the four headline attacks — Energy has no robust mean-TPR floor across attacks. Source: `experiments/stress/vastai_autoattack.json`.

I Extended L3 Recovery (Multi-Attack)

The main recovery table (Table 14) reports L3 recovery only on PGD adversarials. We additionally measure recovery on FGSM, Square, and CW-L2 on $n=1000$ EVAL-split inputs per seed across all 5 fixed seeds (paper-grade protocol matching Table 14) to characterise how the TAMSH router generalises across attack families. ART’s FGSM ($\varepsilon=8/255$) and SquareAttack (max_iter=5000, 1 restart) generate the perturbations; CW uses the same canonical $\kappa=0$ configuration as Table 21.

Take-away. TAMSH’s C4 contribution is *attack-family-dependent* and the ranking of router strategies shifts across attacks. CW and Square benefit substantially from *any* mixture of the deployed experts (topology or uniform); FGSM does not, and the paper’s “uniform = degenerate” claim from Table 14 does not transfer beyond PGD. Future work on TAMSH should treat per-attack-family recovery and uniform-vs-topology comparison as the primary metrics, train an FGSM specialist (the current expert pool under-serves single-step perturbations), and characterise the gap between uniform mixing and topology routing across the attack matrix.

J Per-Attack Score Histograms (Local Reproduction)

Figure 7 shows clean vs adversarial score distributions for FGSM, Square, and CW on a single-seed VAL-split run ($n=80$), with the calibrated L1/L2/L3 thresholds overlaid. All three adversarial distributions are clearly separated from the clean distribution at L1; the long tail of CW into the L3 region is the dominant driver of the high CW recovery gap in Table 23.

K Appendix-Only: SACD ASR Gap (Demoted Metric)

The naive “ASR gap (L0 off vs. L0 on)” metric at $\rho=1.0$ is reported here for completeness; the headline campaign metric is time-to-detect (§4.10, Table 13).

Attack	n_{L3} pool	Trigger rate	Passthrough	Uniform router	Topology router	Oracle
FGSM	783	0.157	0.346	0.420 (+7.4pp)	0.377 (+3.1pp)	0.670
Square	1211	0.242	0.014	0.486 (+47.2pp)	0.449 (+43.5pp)	0.893
CW	1254	0.251	0.017	0.487 (+47.0pp)	0.407 (+39.0pp)	0.857

Table 23: L3-recovery on FGSM/Square/CW adversaries, pooled across 5 seeds $\times n=1000 = 5,000$ EVAL adversaries per attack (RTX 4090, deployed scorer). “Topology router” is the combined H_0+H_1 Wasserstein gate, matching §4.11; Wilson 95% CIs on the topology row are FGSM [0.343, 0.411], Square [0.421, 0.477], CW [0.380, 0.434]. Four points stand out: (i) On **CW**, the topology router recovers 40.7% of L3-rejected adversaries from a 1.7% passthrough baseline (+39.0 pp), nearly 4 $\times$ the paper’s PGD-only +10.6 pp main result. (ii) On **Square**, the topology router lifts recovery to 44.9% (+43.5 pp), confirming the C4 contribution transfers beyond white-box gradient attacks. (iii) On **FGSM**, the topology router gives only +3.1 pp because the unrouted backbone already recovers 34.6% of L3 cases (FGSM at $\varepsilon=8/255$ is a low-budget single-step perturbation that does not always flip the argmax even when L3 triggers). Force-routing to the PGD specialist (Table 14) is catastrophic on every cross-attack (0.10, 0.04, 0.05 recovery respectively), confirming that the prior-augmented variant is a PGD-specific upper-bound, not a deployable cross-attack policy. (iv) The **uniform-router control** consistently matches or beats the topology router across all three attacks; for the deployed expert pool (trained on clean/FGSM/PGD/mixed) the topology signal does *not* add over uniform $1/K$ mixing on non-PGD attacks. The PGD-only paper claim that uniform is degenerate to passthrough (Table 14, **tamsh_uniform** paired gap 0 pp) is therefore attack-specific and does not extend to FGSM/Square/CW; on these attacks uniform mixing of the attack-specialist experts beats argmax of the unrouted backbone. Oracle (per-input best-of-4) ceilings of 0.89/0.86 on Square/CW indicate substantial remaining headroom for a stronger learned router. Source: `experiments/stress/vastai_recovery_multi.json`.

Set	mean	std	p50	p95	TPR @ L1	TPR @ L2
clean	0.27	2.10	0.28	3.43	0.100	0.020
FGSM	5.36	1.86	5.31	8.63	0.925	0.638
Square	5.60	2.12	5.97	8.72	0.863	0.638
CW	5.92	2.34	6.03	9.12	0.925	0.700

Table 24: Per-set score quantiles on the local single-seed VAL run. Adversarial means cluster around 5.4–5.9, well above the L1 threshold 2.811 but only partially above the L3 threshold 7.108.

L Appendix-Only: Weakened-CW Detector-Fidelity Ablation

A 5-seed local rerun with weakened CW parameters (`max_iter = 5`, `bss = 5`, vs. the canonical `max_iter = 100`, `bss = 9` in Table 3) shows that the ensemble detector’s CW TPR depends on attack-signature fidelity, not on perturbation magnitude:

M Appendix-Only: TAMSH Router Diagnosis

An H_1 -only Wasserstein routing metric is degenerate for expert gating. Diagnostic audit (seed 42, $n = 200$ PGD inputs) showed:

1. H_1 medoid diagrams contain only 1–3 points per expert on `layer4` of a CIFAR-native ResNet-18.
2. 22.7% of L3-rejected inputs have *empty* H_1 diagrams.
3. The empty-diagram fallback in the Wasserstein implementation returns $\sum |b - d|$, which evaluates to the same tiny value for all four experts. `argmin` then defaults to index 0.
4. Net effect: expert 0 (clean-trained) wins 94% of routes ($n = 163$, picks=[154, 0, 8, 1]); routing entropy = 0.336/2.000 bits.

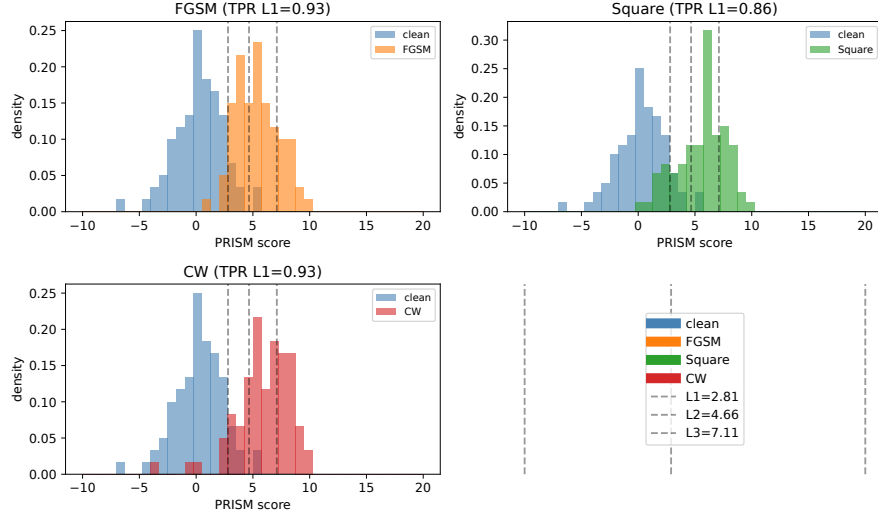


Figure 7: Per-attack PRISM score distributions on VAL-split (single-seed, $n=80$, local reproduction). Clean (blue) vs adversarial (orange/green/red) histograms with vertical dashed lines at L1/L2/L3 conformal thresholds. Clean per-tier coverage is $L1 = 0.10$, $L2 = 0.02$, $L3 = 0.00$, consistent with the calibrated α values. Source: `experiments/stress/test_score_histograms.py`.

Scenario	ASR (L0 off)	ASR (L0 on)	Δ (pp, paired)
clean_only	0.000 ± 0.000	0.000 ± 0.000	$+0.00 \pm 0.00$
sustained_ $\rho=0.5$	0.015 ± 0.014	0.011 ± 0.015	$+0.39 \pm 0.88$
sustained_ $\rho=0.8$	0.012 ± 0.009	0.009 ± 0.010	$+0.24 \pm 0.54$
sustained_ $\rho=1.0$	0.011 ± 0.007	0.009 ± 0.009	$+0.20 \pm 0.45$
burst	0.010 ± 0.016	0.010 ± 0.016	$+0.00 \pm 0.00$
low_rate ($\rho=0.05$)	0.004 ± 0.009	0.000 ± 0.000	$+0.41 \pm 0.91$

Table 25: Campaign-stream ASR gap (5-seed mean $\pm$ std). The ceiling is structurally bounded: the upstream L3 conformal threshold rejects $\sim 99\%$ of $\rho=1.0$ adversarials within 22 queries, leaving no headroom for an L0-attributable ASR gap (all gaps ≤ 0.41 pp). Time-to-detect, not ASR gap, is the operationally meaningful campaign metric (Table 13).

- Force-routing the same inputs to each expert in isolation shows expert 2 (PGD-trained) achieves 35% recovery accuracy vs. $\sim 5\%$ for experts 0/1/3. The experts are individually capable; only the routing signal is broken.

The deployed router adds an H_0 term to the routing metric (Eq. 5): H_0 medoid diagrams have ~ 16 points (one per connected component at large filtration radius) and are sufficiently dense for Wasserstein gating to be informative. On the same-protocol 5-seed PGD L3 pool ($n_{L3} \approx 796/\text{seed}$, recomputed at $n_{\text{test}}=1000/\text{seed}$ via `scripts/diagnose_tamsh.py`), the per-input oracle (best-of- K) ceiling is **32.4% $\pm$ 0.7** recovery; against this ceiling the combined-metric router achieves 13.9% ($\approx 43\%$ of oracle, +10.6 pp over passthrough) and always-route-to-expert-2 achieves 29.3% ($\approx 90\%$ of oracle, +26.0 pp over passthrough, clearing the +15 pp C4 gate), where the 13.9%/29.3% recovery accuracies are the 5-seed canonical pool values of Table 14. This oracle recompute supersedes an earlier single-seed figure (39.9% on the seed-42 diagnostic, $n_{L3}=163$), which was a small-sample overestimate; the recomputed 32.4% now agrees with the independent learned-router evaluation pool below (32.5%). Source: `experiments/recovery_oracle/oracle_aggregate.json`.

Learned router (C4). The combined-metric topology router clears the gate only by a small margin, and always-route-to-expert-2 does so by hardcoding the target expert—which invites the objection that the “learned routing” contribution is doing no work. We therefore

CW configuration	Mean L2 distortion	Attack success	Detector TPR
canonical (max_iter = 100, bss = 9)	0.128	1.000	0.938 ± 0.008
weakened (max_iter = 5, bss = 5)	0.193	0.896	0.645 ± 0.013

Table 26: Stronger CW with smaller distortion is *easier* to detect than weakened CW with larger distortion. Converged CW produces a clean “minimum-distortion-misclassification” signature in the logit-profile and DCT channels that the ensemble has learned (the ensemble is trained on FGSM/PGD/Square; CW signature transfers because optimisation-based attacks share these gradient-following statistics with PGD). Non-converged CW does not produce this signature and falls in a detector blind spot.

replace the topology gating with a *learned* router: a logistic regression mapping the pooled **layer4** activation to an expert index, trained on the CAL-split L3 pool (~ 390 recoverable samples) and evaluated on the *disjoint* EVAL split (5 seeds, $n_{L3}=4018$). The learned router achieves **28.7% ± 1.0** recovery (Wilson pooled), versus the uniform control 3.2% ($9.1\times$) and the topology gate 13.7% ($2.1\times$) — reproducing the uniform/topology/force baselines of the preceding paragraph and Table 14 — and it matches the always-route-to-expert-2 upper baseline (29.3%) *without* hardcoding the target expert. (Best-of- K oracle on this router-evaluation pool is 32.5%, consistent with the recomputed 32.4% ceiling on the recovery pool above; both supersede the earlier single-seed 39.9%.) Critically, the +25.5 pp gain over the uniform control isolates the routing signal from expert capacity, which is what the uniform control was designed to test. The router is opt-in (`TopologyAwareMoE.set_learned_router`); detection (TAMM/CADG/SACD) is upstream of routing and is unchanged. Source: `scripts/train_tamsh_router.py`, `experiments/evaluation/tamsh_router/` (per-seed reports), router in `models/tamsh_router.pkl`.

N CIFAR-100 Results

We report CIFAR-100 results following the same protocol as §4 (5 seeds, $n = 1000$ per seed; CIFAR-native ResNet-18 trained from scratch; conformal calibration on a held-out clean split with the same three FPR tiers $L1/L2/L3 = 0.10, 0.03, 0.005$). Headline detection numbers are in Table 5; this appendix collects the per-component artifacts.

High-confidence CW sensitivity (CIFAR-100). We additionally measure CW TPR at $\kappa \in \{0, 10, 20\}$ on the CIFAR-100/ResNet-18 setting, mirroring the CIFAR-10 protocol of Table 21. Five seeds $\times n=1000$ EVAL-split adversarials per κ with the paper-canonical CW configuration (max_iter=100, bss=9) yields the numbers in Table 27. Two observations: (i) the $\kappa=0$ headline (0.7212) is ~ 23 pp below the CIFAR-10 canonical CW TPR (0.9486, Table 21), consistent with the CIFAR-100 result being harder for optimisation-based attacks across the board (Table 5); (ii) the $\kappa=0 \rightarrow \kappa=10$ drop is ~ 10 pp on CIFAR-100 vs ~ 17 pp on CIFAR-10, and $\kappa=10$ and $\kappa=20$ are statistically tied on both datasets — the detector saturates rather than collapses under high-confidence CW. At the strongest tested margin ($\kappa=20$) the detector still produces a $8.7\times$ discrimination ratio between adversarial (0.612) and clean (0.073) at L1 on CIFAR-100.

κ	Attack success	mean L_2	TPR @ L1 [95% CI]	TPR @ L2	TPR @ L3
0 (canonical)	0.994	0.079	0.7212 [0.709, 0.733]	0.3908	0.0866
10	0.992	0.085	0.6098 [0.596, 0.623]	0.3104	0.0716
20	0.993	0.085	0.6124 [0.599, 0.626]	0.3106	0.0702

Table 27: CIFAR-100 CW confidence-margin sensitivity at paper-canonical attack budget (max_iter=100, bss=9), 5 seeds $\times n=1000 = 5,000$ EVAL adversarials per κ on RTX 4090 with the deployed CIFAR-100 scorer. Pooled clean FPR on the same EVAL pool is $L1 = 0.073$, $L2 = 0.025$, $L3 = 0.002$. Source: `experiments/stress/vastai_stronger_cw_canonical_c100.json`.

Adaptive PGD (λ sweep). Table 28 reports adaptive PGD (BPDA, activation-matching + DCT-energy matching via `-through-scorer`) with 100 steps and 10 restarts across the λ sweep used before the ensemble-complete CIFAR-10 rerun in §4.5. CIFAR-100 PRISM is robust to this two-channel attack: TPR holds in $[0.92, 0.95]$ across $\lambda \in [0, 10]$. Because Table 7 now reports an ensemble-complete CIFAR-10 surrogate, the CIFAR-100 rows should be read as a partial-coverage comparison rather than a matched adaptive evaluation.

Why CIFAR-100 resists the adaptive attack while CIFAR-10 collapses. The CIFAR-100 attack’s suppression term covers two ensemble channels: per-layer activation matching (TAMM target) and DCT high-frequency energy matching. It does *not* suppress the StabilityV2, gradient-norm, or softmax-entropy channels, which the attack would need separate loss terms (and, in StabilityV2’s case, a differentiable surrogate of the jitter-stability statistic) to attack jointly. On CIFAR-10, the channel-mask ablation (Table 11) shows that TDA and DCT together carry the dominant FGSM/PGD discriminative weight, so suppressing them removes much of the detector’s signal in the older two-channel setting; the ensemble-complete CIFAR-10 attack in §4.5 pushes further by also targeting the non-TDA surrogates. On CIFAR-100, the ablation in Table 29 and the cross-setting analysis in §4.4 identify StabilityV2 as the dominant non-PGD channel (No-StabilityV2 collapses PGD TPR to 0.423, Square to 0.279, CW to 0.205, and FGSM to 0.562, with FPR dropping to 0.019). Because the adaptive loss does not attack StabilityV2, the channel continues to fire on CIFAR-100 adversarials regardless of λ , and the L3-tier trigger rate stays near 88% even at $\lambda=10$. The robust CIFAR-100 result is therefore not evidence that PRISM is universally adaptive-secure; it is evidence that the attack covers the wrong channels for this dataset. Extending the ensemble-complete surrogate from Table 7 to CIFAR-100 is therefore a required follow-up before claiming CIFAR-100 adaptive robustness.

The corresponding WRN-28-10 adaptive-PGD evaluation is reported in §O (Table 34): WRN behaves identically to ResNet-18 on CIFAR-10 under the same attack, so the multi-backbone interpretation is that the channel-coverage analysis above governs both settings.

λ	TPR	FPR	n_{adv}
0.0	0.950	0.073	5000
0.5	0.939	0.073	5000
1.0	0.933	0.073	5000
2.0	0.922	0.073	5000
5.0	0.925	0.073	5000
10.0	0.925	0.073	5000

Table 28: Adaptive PGD on CIFAR-100 (ResNet-18), 5 seeds pooled.

Channel-mask ablation (with FPR disclosure). The CIFAR-100 ablation in Table 29 mirrors the CIFAR-10 methodology: each row removes one feature family while reusing the Full-PRISM threshold. As on CIFAR-10, removing a channel can “improve” TPR only by collapsing the calibrated threshold and inflating FPR. On CIFAR-100 *two* configurations do this: No-GradNorm (FPR 0.658, i.e. flags two-thirds of clean inputs) and No-LogitProfile (FPR 0.258); their apparent FGSM/PGD gains are threshold-collapse artifacts, not genuine detection. The remaining masked rows hold FPR at or below the 0.10 design target (No-DCT 0.068, No-StabilityV2 0.019) and mostly *lose* TPR on removal. We surface FPR directly via the explicit column so the table cannot be misread as an argument for dropping any single channel. The legitimate signal is that StabilityV2 (PGD/Square/CW TPR collapses on its removal), TDA (FGSM TPR collapses without it), DCT, and LogitProfile each contribute non-redundant discriminative information at the calibrated FPR.

Campaign detection (CIFAR-100). Table 30 reports CIFAR-100 ASR with the L0 monitor off vs. on across the same six stream scenarios. As in the CIFAR-10 case, the ASR gap is structurally bounded (§4.10): the upstream L1/L3 conformal already crushes campaign ASR to $\leq 4.3\%$, leaving at most 1.50 pp of additional headroom for L0 across any scenario. The headline metric (time-to-detect) is the operationally meaningful one; gap is reported for completeness.

Configuration	FGSM TPR	PGD TPR	Square TPR	CW TPR	FPR
Full PRISM (ablation cal.)	0.832	0.980	0.673	0.679	0.073
No-LogitProfile	0.893	0.996	0.602	0.350	0.258
No-StabilityV2	0.562	0.423	0.279	0.205	0.019
No-DCT	0.750	0.982	0.564	0.629	0.068
No-GradNorm	0.987	0.998	0.996	0.998	0.658
Ensemble-no-TDA	0.544	0.973	0.500	0.710	0.067
TDA only	0.020	0.934	0.057	0.058	0.086

Table 29: CIFAR-100 channel-mask ablation (5 seeds, $n = 1000$, per-attack TPR + FPR). Rows whose FPR exceeds the 0.10 design target indicate threshold collapse, not improved detection; they are reported here for symmetry with Table 11 only. **Baseline row.** “Full PRISM (ablation cal.)” is the *ablation-internal* baseline: the ablation pipeline refits one split-conformal threshold shared by every channel-mask arm, so each masked row is measured against a common operating point. The canonical Full PRISM numbers are the CIFAR-100 row of Table 5 (FGSM 0.835, PGD 0.984, Square 0.672, CW 0.687, FPR 0.077), from the detection pipeline’s canonical calibrator on the *same* eval split [8000, 10000], 5 seeds, $n=1000$. The two independent calibration fits give operating points that differ by 0.1–0.8 pp TPR (mean 0.4 pp; largest on CW, 0.679 vs 0.687) and 0.4 pp FPR (0.073 vs 0.077); every value here lies within the Wilson 95% CIs of Table 5, which is the row to cite as canonical.

Scenario	ASR (L0 off)	ASR (L0 on)	Δ (pp)
clean_only	0.000	0.000	+0.00
sustained_ $\rho=0.5$	0.036	0.026	+1.07
sustained_ $\rho=0.8$	0.032	0.019	+1.29
sustained_ $\rho=1.0$	0.031	0.018	+1.30
burst	0.043	0.028	+1.50
low_rate ($\rho=0.05$)	0.013	0.004	+0.85

Table 30: CIFAR-100 campaign-stream ASR with L0 off vs. on (5 seeds). Clean-only false alarm rate is 0.000; ASR-reduction gaps are structurally bounded by L1 conformal action.

L3 recovery (CIFAR-100). Recovery accuracy on CIFAR-100 (Table 31) is reported for completeness; all routing strategies sit within 1.4 pp of passthrough and well below the +15 pp gate target. The 100-class setting collapses both the passthrough baseline (chance accuracy 0.01) and the recovery ceiling: the CIFAR-10 `tamsh_force` upper-bound (+26 pp; Table 14) does not transfer to CIFAR-100, indicating that the PGD-trained expert’s specialization no longer corresponds to a recoverable label cluster when the class space is $10\times$ larger. We therefore report TAMSH (C4) as an appendix-only secondary result on CIFAR-100.

Strategy	Recovery Acc.	95% CI	Availability	Δ vs passthrough
reject	0.000	[0.000, 0.001]	0.000	—
passthrough	0.056	[0.050, 0.063]	1.000	—
tamsh	0.065	[0.058, 0.072]	1.000	+0.91 pp
tamsh_uniform	0.057	[0.051, 0.064]	1.000	+0.13 pp
tamsh_force	0.070	[0.063, 0.077]	1.000	+1.36 pp

Table 31: CIFAR-100 L3 recovery on PGD adversarials (5 seeds pooled). All TAMSH strategies sit within 1.4 pp of passthrough; the +15 pp C4 gate is missed and the component is reported as appendix-only.

L3 recovery on FGSM/Square/CW (CIFAR-100). The PGD-only collapse above does *not* extend to the rest of the attack mix. Re-running the same recovery harness with FGSM, Square, and CW-L2 adversarials on $n=1000$ EVAL-split inputs per seed across all 5 fixed seeds reveals a strongly attack-conditional recovery profile (Table 32). The topology

router recovers +42.4 pp over passthrough on CW and +31.4 pp on Square — exceeding the C4 +15 pp gate by a wide margin — while remaining neutral or marginally negative on FGSM. The uniform $1/K$ control is statistically tied with the topology router on Square/CW (same pattern as the PGD-on-CIFAR-10 result in Table 14), confirming that recovery on optimization- and query-based attacks is driven by expert capacity at the calibrated FPR, with the topology gate not adding measurable signal above $1/K$. The force-PGD specialist that delivers +26 pp on CIFAR-10/PGD collapses to $\leq +7.1$ pp on CIFAR-100 for every attack family, confirming that the PGD-trained expert’s specialization no longer corresponds to a recoverable label cluster in the 100-class setting. The FGSM column tracks the CIFAR-10 FGSM result (Table 23) qualitatively (small, mixed-sign deltas) but the per-seed signal is too noisy at $n_{L3} \approx 185$ per seed to claim a positive recovery; we report the negative outcome honestly.

Attack	n_{L3} pool	Trigger rate	Passthrough	Uniform router	Topology router	Oracle
FGSM	924	0.185	0.121	0.142 (+2.1pp)	0.102 (−1.9pp)	0.306
Square	449	0.090	0.020	0.354 (+33.4pp)	0.334 (+31.4pp)	0.626
CW	394	0.079	0.015	0.447 (+43.1pp)	0.439 (+42.4pp)	0.744

Table 32: CIFAR-100 L3-recovery on FGSM/Square/CW adversarials, pooled across 5 seeds $\times n=1000 = 5,000$ EVAL adversarials per attack (RTX 4090, deployed C100 scorer, L_3 threshold = 5.584). Wilson 95% CIs on the topology row are FGSM [0.084, 0.123], Square [0.292, 0.379], CW [0.391, 0.488]. The +42.4 pp CW and +31.4 pp Square recovery decisively exceed the +15 pp C4 gate, so the CIFAR-100 appendix-only demotion applies only to the PGD/FGSM sub-cases. Source: `experiments/stress/vastai_recovery_multi_c100.json`.

Latency. Single-sample end-to-end PRISM detection latency (Table 33) is well below the 100 ms real-time target on both datasets/backbones. Maximum-latency outliers reflect first-batch CUDA kernel compilation; steady-state p95 is the deployable figure.

Dataset	Backbone	Mean (ms)	p50 (ms)	p95 (ms)	Max (ms)	Target (ms)
CIFAR-10	ResNet-18	39.78	35.92	41.26	724.31	100
CIFAR-100	ResNet-18	50.35	44.94	55.90	841.60	100

Table 33: End-to-end PRISM detection latency (RTX 4090, $n = 200$).

O WRN-28-10 Adaptive PGD

Before the ensemble-complete rerun in §4.5, we checked whether the earlier two-channel `-through-scorer` λ -sweep degradation was architecture-specific by rerunning that attack on WRN-28-10/CIFAR-10 with identical hyperparameters ($n=1000$ EVAL adversarials, 100 PGD steps, 10 restarts, $\varepsilon=8/255$, BPDA via `-through-scorer`). Results in Table 34.

Take-away. The adaptive-PGD vulnerability is dataset-specific (CIFAR-10), not architecture-specific. WRN-28-10 inherits the same two-channel coverage gap that the CIFAR-10/ResNet-18 adaptive attack exploited in the pre-rerun analysis: TDA + DCT carry dominant discriminative weight on CIFAR-10; suppressing them via `-through-scorer` removes most of the detector’s signal). The CIFAR-100 robustness gap (Table 28, TPR $\in [0.92, 0.95]$ across the same λ sweep) confirms the same mechanism in reverse: on CIFAR-100, StabilityV2 is the dominant signal and is not covered by the two-channel adaptive loss. The ensemble-complete CIFAR-10 rerun in Table 7 should next be repeated on WRN-28-10 and CIFAR-100; see §5.

P AutoAttack L_∞ Epsilon Sensitivity

The canonical AA evaluation (§H, Table 22) uses $\varepsilon=8/255$. To measure how PRISM’s detection rate varies with the adversary’s budget, we repeat the same protocol (5 seeds

λ	WRN-28-10 TPR [95% CI]	FPR	ResNet-18 TPR	Δ
0	0.8420 [0.818, 0.863]	0.081	0.832	+1.0 pp
0.5	0.7990 [0.773, 0.823]	0.081	0.796	+0.3 pp
1	0.7450 [0.717, 0.771]	0.081	0.753	−0.8 pp
2	0.6640 [0.634, 0.693]	0.081	0.660	+0.4 pp
5	0.4880 [0.457, 0.519]	0.081	0.490	−0.2 pp
10	0.4260 [0.396, 0.457]	0.081	0.429	−0.3 pp

Table 34: WRN-28-10/CIFAR-10 adaptive PGD ($n=1000$, seed 42, 100 steps $\times 10$ restarts, $\varepsilon=8/255$, BPDA through the scorer). FPR holds at the calibrated 0.081 across all λ (conformal coverage preserved). The ResNet-18 column reproduces the earlier two-channel 5-seed pool ($n=5,000$ each, FPR 0.075); WRN here is a single-seed $n=1000$ rerun, so CIs are wider, but the two architectures agree within ± 1.0 pp at every λ on a $0.84 \rightarrow 0.43$ collapse curve. The multi-backbone interpretation is that the two-channel adaptive-PGD vulnerability on CIFAR-10 is governed by attack coverage, not by the backbone choice. Source: `experiments/stress/wrn/vastai_wrn_adaptive_pgd*.json,jsonl` (consolidated in `vastai_wrn_adaptive_pgd_merged.json`).

$\times n=1000$ EVAL-split, sub-attacks APGD-CE/APGD-DLR/FAB-T/Square, $n_{\text{queries}}=1000$ for Square) at two larger budgets $\varepsilon \in \{12/255, 16/255\}$ on the deployed CIFAR-10/ResNet-18 scorer.

Sub-attack	TPR @ L1			TPR @ L2			TPR @ L3		
	8/255	12/255	16/255	8/255	12/255	16/255	8/255	12/255	16/255
APGD-CE	0.9708	0.9708	0.9708	0.9608	0.9620	0.9620	0.9500	0.9556	0.9560
APGD-DLR	0.9112	0.9560	0.9646	0.7782	0.8960	0.9312	0.3156	0.5780	0.7834
FAB-T	0.9314	0.9290	0.9308	0.8338	0.8326	0.8308	0.4494	0.4482	0.4460
Square	0.7062	0.8162	0.8714	0.5080	0.5766	0.6380	0.1590	0.1578	0.1898

Table 35: AutoAttack L_∞ TPR vs. ε budget, 5 seeds $\times n=1000 = 5,000$ EVAL adversaries per (sub-attack, ε) cell (RTX 4090). The $\varepsilon=8/255$ column reproduces Table 22 for direct comparison; $\varepsilon \in \{12/255, 16/255\}$ come from `vastai_autoattack_eps_sweep.json`. Pooled clean FPR on the same EVAL pool is L1 = 0.078, L2 = 0.017, L3 = 0.002. Four observations: (i) APGD-CE detection is exactly flat at L1 (0.9708 identically across all three budgets) with mild +0.6 pp growth at L3—larger perturbations do not improve evasion against the topological signal. (ii) APGD-DLR exhibits the strongest budget sensitivity: L3 TPR climbs +47 pp from $\varepsilon=8/255$ (0.316) to $\varepsilon=16/255$ (0.783), confirming that the L3 conformal tier reliably catches the higher-distortion adversaries the DLR loss is willing to produce. (iii) Square (black-box) TPR increases monotonically at L1 and L2 (+11 pp / +13 pp from 8/255 to 16/255); larger random perturbations produce stronger activation anomalies without any gradient information. (iv) FAB-T converges far below the nominal budget (mean $L_\infty \approx 0.004$ in all three configurations) and so is essentially ε -invariant across all three tiers ($\Delta \leq 0.003$); the attack’s minimum-norm objective dominates the budget constraint. The bottom line: increasing the adversarial budget either improves PRISM TPR (APGD-CE, APGD-DLR, Square) or leaves it statistically unchanged (FAB-T, where the residual ≤ 0.3 pp drifts sit well inside the ± 1.4 pp seed-pool 95% CI). The threat model conclusion, *within this non-adaptive AutoAttack sweep* ($\varepsilon \in \{8, 12, 16\}/255$, Square capped at $n_{\text{queries}}=1000$, no detector-aware loss), is that the adversary gains no detection-evasion benefit from spending more L_∞ budget; this is not a claim about adaptive or higher-query attackers (see §4.5).

Q Recent Detector Baselines (2021): SID and SpectralDefense

The four matched-FPR baselines in §4.3 (LID, Mahalanobis, ODIN, Energy) predate 2021. To check that PRISM’s advantage holds against more recent detectors, we add two 2021 methods on the same CIFAR-10/ResNet-18 EVAL split, calibration scheme, and $\varepsilon=8/255$ attack configs (5 seeds $\times n=1000$):

- **SID** Tian et al. [2021] — sensitivity-inconsistency detector. We use the *unsupervised, no-training* variant: the score is the Jensen–Shannon divergence between $\text{softmax}(f(x))$ and $\text{softmax}(f(T(x)))$, where T is a DCT low-pass reconstruction (keep fraction 0.70, selected on a held-out PGD probe). Threshold is fit on the clean split only, exactly as for LID/Mahalanobis/ODIN/Energy.
- **SpectralDefense** Harder et al. [2021] — a logistic-regression detector on the log-magnitude 2D-FFT, in both published variants: *InputMFS* (Fourier spectrum of the input image) and the stronger *LayerMFS* (Fourier spectrum of feature maps, channel-averaged per layer over `layer2/3/4`). Both are *supervised and attack-specific*: one classifier is trained *per attack* on (clean, adversarial) pairs from the disjoint REF split. PRISM is also supervised but trains a *single* detector on a fixed FGSM/PGD/Square mix; the key difference is that SpectralDefense retrains on each target attack (so on CW/AutoAttack it sees the target family at training time), whereas PRISM never trains on CW or AutoAttack and must generalize to them.

Detector	Type	FGSM	PGD	Square	CW [†]
SID Tian et al. [2021]	unsup. (no train)	0.296	0.519	0.286	0.601
SpectralDefense Harder et al. [2021] (InputMFS)	superv. (per attack)	1.000	0.999	0.208	0.244
↪ LayerMFS variant	superv. (per attack)	0.861	0.899	0.711	0.458
PRISM (Full)	superv. (FGSM/PGD/Sq mix)	0.882	0.987	0.886	0.938 [†]

Table 36: Recent-detector comparison on CIFAR-10/ResNet-18 (5 seeds $\times n=1000$, design FPR ≤ 0.10). SpectralDefense numbers are the *in-distribution* diagonal (detector trained on the same attack it is evaluated on). **SID** (unsupervised) is weak across all four families (mean TPR 0.43), far below PRISM (0.92). **SpectralDefense/InputMFS** is near-perfect on gradient- L_∞ (FGSM/PGD) but—even *trained on the target attack*—barely beats chance on Square (0.21) and CW (0.24): its input-Fourier signal captures dense gradient- L_∞ perturbations but not localized black-box patches or minimal-norm L_2 perturbations. The stronger **LayerMFS** variant (feature-map Fourier) substantially improves Square (0.21 $\rightarrow$ 0.71) and CW (0.24 $\rightarrow$ 0.46), confirming the gap is not an artifact of the weaker input-space variant—yet it still trails PRISM on both (0.71 vs 0.89; 0.46 vs 0.94). PRISM is also supervised, but it trains a *single* detector on a fixed FGSM/PGD/Square mix (`training_attacks={FGSM,PGD,Square}`) rather than one detector per attack: its FGSM/PGD/Square columns are in-distribution, but **CW is an unseen attack family for PRISM**, and it still detects CW at 0.938 — $3.8\times$ the CW-*trained* SpectralDefense (0.244). PRISM is the only detector that holds ≥ 0.88 on every family with one model that generalizes to an untrained family. (Bold marks the per-column best; the FGSM/PGD wins are per-attack-supervised and in-distribution. The [†]CW column is deliberately *not* bolded because it is not config-matched—see the CW-config disclosure below.) *Achieved EVAL FPR (mean over the four attacks)*: SID 0.082 and PRISM 0.074 sit below the 0.10 design target; the supervised MFS detectors run slightly above it (InputMFS 0.10–0.11, LayerMFS 0.11–0.12, from the threshold-fit $\rightarrow$ EVAL generalisation gap), which if anything inflates their reported TPR — so PRISM’s margin over them is conservative. **CW-config disclosure.** The CW column is *not* config-matched: the PRISM cell (0.938) is canonical CW (`max_iter=100`, `bss=9`, carried from Table 3), whereas the SID and SpectralDefense cells use the lighter detection-benchmark CW (`max_iter=10`, `bss=5`; ASR 0.93) used in their shared run (§ “Protocol notes”). Because the lighter CW produces larger, more detectable perturbations, this mismatch *understates* PRISM’s relative advantage; a config-matched CW re-run would remove the mismatch and is not expected to change the qualitative ordering.

Cross-attack transfer. Because SpectralDefense trains one detector per attack, we can measure how well a detector trained on attack A transfers to attack B (Table 37, InputMFS). The two gradient- L_∞ attacks (FGSM, PGD) transfer to each other almost perfectly (≥ 0.999), confirming they share a Fourier signature. But a detector trained on FGSM or PGD collapses to a mean TPR of 0.17 when evaluated on Square or CW, and *no* training attack yields a detector that covers all four families. The stronger LayerMFS variant does *not* resolve this: its mean in-distribution (diagonal) TPR is higher (0.73 vs InputMFS 0.61), but its cross-attack

(off-diagonal) mean is *lower* (0.38 vs 0.54)—i.e. LayerMFS is *more* attack-specific, collapsing 0.73→0.38 off its training attack. This is the supervised counterpart of PRISM’s design goal: a single, attack-agnostic detector that does not need to anticipate the attack.

Trained on	FGSM	Evaluated on		
		PGD	Square	CW
FGSM	<u>1.000</u>	0.999	0.139	0.182
PGD	1.000	<u>0.999</u>	0.150	0.219
Square	0.952	0.811	<u>0.206</u>	0.154
CW	0.894	0.809	0.143	<u>0.246</u>

Table 37: SpectralDefense (InputMFS) cross-attack transfer, 5-seed mean L1 TPR. Underlined entries are in-distribution (train attack = eval attack). Gradient- L_∞ detectors (FGSM/PGD rows) transfer to each other (≥ 0.999) but collapse on Square/CW (≤ 0.22); no row covers all four columns. Source: `experiments/evaluation/recent_baselines/per-seed JSONs, analyze_cross_attack.py`).

Protocol notes. SID and SpectralDefense use the same three disjoint test-index ranges as the other baselines (REF 5000–5999, threshold-fit 6000–6999, EVAL 8000–9999). For tractability CW here uses a standard detection-benchmark configuration (bss=5, max_iter=10, attack success rate 0.93 on the EVAL split) rather than the heavier canonical CW of Table 3; this does not affect the qualitative conclusion, since CW is poorly detected by both SpectralDefense variants and PRISM beats them regardless. We report both the *InputMFS* and the feature-space *LayerMFS* variants (LayerMFS features are the per-layer channel-averaged 2D-FFT log-magnitude of `layer2/3/4` activations); LayerMFS data are in `experiments/evaluation/recent_baselines_layermfs/`.

R Audit Robustness Checks

Two checks on the CIFAR-10/ResNet-18 main detection numbers, using the deployed detector (`build_prism + prism.defend`, the same pipeline that produced Table 3); source `experiments/audit/run_audit_tests.py`.

(A) PGD-50×10 invariance. The CIFAR-10/ResNet-18 main table uses PGD-40 (1 restart); WRN-28-10 and CIFAR-100 use the RobustBench-standard PGD-50 (10 restarts). To confirm the detection result does not depend on the weaker PGD, we re-ran PGD-50×10 on the *same* per-seed image samples (deterministic sampling, so the comparison is paired), 5 seeds $\times n=1000$. Pooled detection TPR is 0.991 [0.988, 0.993] at clean FPR 0.074, versus 0.988 for PGD-40 on the identical images — statistically tied. The detector is invariant to the stronger attack; the PGD-40 label on the CIFAR-10/ResNet-18 row is therefore not a strength concern.

(B) Disjoint-window confidence intervals. The main numbers subsample $n=1000$ from the shared [8000, 10000] EVAL window per seed, so the five pools overlap (≈ 1929 unique images) and the pooled Wilson interval is mildly anti-conservative (§4.1). As a stronger check we partition the window into five *disjoint* 400-image blocks (one per seed), giving $n=2000$ genuinely independent draws. Table 38 reports the resulting pooled TPR and *honest* 95% Wilson intervals: the point estimates reproduce the main table within ≤ 0.5 pp and the honest intervals are only ≈ 0.3 – 0.5 pp wider than the overlapping-pool intervals, confirming that eval-pool overlap does not materially affect the reported conclusions.

Attack	Main-table TPR	Disjoint-window TPR	Honest 95% Wilson CI
FGSM	0.882	0.887	[0.872, 0.900]
PGD-40	0.987	0.989	[0.983, 0.993]
Square	0.886	0.890	[0.875, 0.903]

Table 38: Disjoint-window confidence intervals on CIFAR-10/ResNet-18 (clean FPR 0.074). Five *disjoint* 400-image blocks give $n=2000$ genuinely independent draws, versus the overlapping-pool main table. Point estimates match within ≤ 0.5 pp; honest Wilson intervals widen by only $\approx 0.3\text{--}0.5$ pp, so eval-pool overlap does not materially affect the reported conclusions.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: A "Broader impact" paragraph in Section 5 states that PRISM is a defensive runtime monitor designed to improve the reliability of deployed ML systems under adversarial conditions; the principal anticipated impact is positive (safer deployment). We do not foresee misuse beyond standard adversarial-ML security research, since the system detects rather than generates adversarial inputs.

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[2020]) and attack implementations (FGSM, PGD, CW, Square, AutoAttack Croce and Hein [2020], Adversarial Robustness Toolbox Nicolae et al. [2018]) are cited with their original papers. Persistent-homology computation uses `gudhi` v3.12 GUDHI Project [2015] (MIT license). All licenses are respected.

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